

Open Science in the Kingdom of Saudi Arabia

National Survey

The 2021 UNESCO Recommendation on Open Science

February 28, 2025

بسم الله الرحمن الرحيم

Table of Contents

ACRONYMS:.....	6
A. GENERAL INFORMATION ABOUT THE RESPONDENT.....	9
A.1 PROJECT TEAM	12
A.2 ACKNOWLEDGMENTS	12
1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE.....	13
1.1. Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?	13
1.2. Is the 2021 Recommendation on Open Science available in the national language(s) of your country?	16
1.3. Have there been awareness-raising activities on open science, including all its key elements, associated benefits, and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities?	16
1.4. Have specific actions been undertaken or are planned to be undertaken by the end of 2025 in your country to incorporate the values and principles of open science ii, in publicly funded research?.....	18
2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE.....	23
2.1 Does your country have a national policy, strategy, or plan of action on science, technology, and innovation (STI)?	23
2.2 In your country, is there a policy, or a set of policies and/or legal frameworks at the national level that address open science or any of its key elements in line with the definition, values, and principles outlined in the 2021 Recommendation on Open Science?	25
2.3 In your country, are there specific policy instruments that aim to promote open science?	38
2.4 In your country, is there a national implementation plan, strategy, policy, or roadmap for the implementation of open science at the national level in line with the 2021 Recommendation?	48
2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at the institutional level, including in the context of research-performing institutions, universities, scientific unions and associations, and learned societies?.....	51
2.6 In your country, are there specific funding mechanisms for open science at the national level?	54

2.7	In your country, have open science practices been integrated into existing research funding mechanisms?	58
2.8	Have efforts been made or are foreseen to be undertaken by the end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation?	62
2.9	Is there a national monitoring framework for open science in your country? Or are there specific indicators of open science included in the national monitoring and evaluation framework for STI policy?	66
3.	INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES	70
3.1.	What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers.....	70
3.2.	What is the latest official data for the percentage of your country's population that has access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers.	70
3.3.	Are there national research and education networks (NRENs) active in the country?	70
3.4.	Are there national or institutional open science infrastructures in your country?	72
3.5.	Does your country host or fund any federated regional or international information technology infrastructure for open science?	75
3.6.	Is your country involved in any North-South, North-South-South, and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional, and national open science platforms?	77
4.	INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY, AND CAPACITY BUILDING FOR OPEN SCIENCE.....	80
4.1	Has systematic capacity building on open science taken place in higher education and/or research-performing institutions in your country?	80
4.2	Have capacity-building programs for policymakers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?	81
4.3	Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025?	82
4.4	Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above?	82

4.5	Have there been any initiatives in your country to support science communication in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers, and the public at large?	83
5.	FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE	85
5.1	Have there been any initiatives at the national or institutional level, e.g. by universities and/or research-performing institutions, to review the existing research assessment and evaluation processes to align them with open science principles and values or are they foreseen by the end of 2025?	85
5.2	Have specific clauses that recognize open science practices, including sharing, collaborating, and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025?	88
5.3	Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025?	89
5.4	Have there been any unintended negative consequences of open science practices reported in your country?	91
6.	PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS	93
6.1	Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process?	93
7.	PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS	103
7.1	How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science?	103
7.2	Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regard to the exchange of best practices, capacity building, and metrics for open science?	104
7.3	Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?	105
8.	GENERAL CONSIDERATIONS	106

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?	106
ANNEX 1: PARTICIPATED ORGANIZATIONS AND CANDIDATES	107

ACRONYMS:

No	Acronyms	Definition
1.	ALECSO	The Arab League Educational, Cultural, and Scientific Organization
2.	CARE	Collective Benefit, Authority to Control, Responsibility, and Ethics
3.	CC-BY	creative Commons Attribution
4.	CLSR	Center of Legal Research and Studies at the Bureau of Experts
5.	Elm	Elm Company
6.	ENERCOMP	Aramco-KAUST Consortium for Non-metallics and Composites in Energy Applications
7.	ETEC	Education & Training Evaluation Commission
8.	FAIR	Findable, Accessible, Interoperable, Reusable
9.	FSC	Federation of Saudi Chambers
10.	GPID	Global Prize for Innovation in Desalination
11.	GRC	Global Research Council
12.	HADITH	The Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith
13.	HRSD	Ministry of Human Resources and Social Development
14.	ICT	Information and Communications Technology
15.	IDS	Prince Saud AL Faisal Institute for Diplomatic Studies
16.	IPA	Institute of Public Administration
17.	IOAP	Institutional Open Access Program
18.	IUOM	Islamic University of Madinah
19.	JazanU	Jazan University
20.	JU	Jouf University
21.	KACARE	King Abdullah City for Atomic and Renewable Energy
22.	KACST	King Abdulaziz City for Science and Technology
23.	KAIMRC	King Abdullah International Medical Research Center
24.	KAPL	King Abdul-Aziz Public Library
25.	KAPSARC	King Abdullah Petroleum Studies and Research Center
26.	KAPSRC	King Abdullah Petroleum Studies and Research Centre
27.	KAUST	King Abdullah University of Science and Technology
28.	KAWLA	King Abdulaziz Complex for Endowment Libraries
29.	KFNL	King Fahad National Library
30.	KFSHRC	King Faisal Specialist Hospital and Research Centre
31.	KFU	King Faisal University
32.	KKU	King Khalid University
33.	KSGAAL	King Salman Global Academy for Arabic Language
34.	KSSO	King Salman Science Oasis
35.	LCGPA	Local Content & Government Procurement Authority

No	Acronyms	Definition
36.	Mawhiba	King Abdulaziz and his Companions Foundation for Giftedness and Creativity
37.	MCIT	The Ministry of Communication and Information Technology
38.	MDPI	Multidisciplinary Digital Publishing Institute
39.	Media	Ministry of Media
40.	MEP	Ministry of Economy and Planning
41.	MEWA	Ministry of Environment, Water and Agriculture
42.	MIM	Ministry of Industry and Mineral Resources
43.	MOE	Ministry of Education
44.	MOENERGY	Ministry of Energy.
45.	NCA	National Cybersecurity Authority
46.	NCSS	National Center for Social Studies
47.	NeLC	National eLearning Center
48.	NEOM	NEOM Education, Research, and Innovation
49.	NOMAD	Novel Methods and Algorithms for Materials Discovery
50.	NREN	National Research and Education Network
51.	NTDP	National Technology Development Program
52.	NTPC	National Transformation Program Center
53.	OECD	Organization for Economic Co-operation and Development
54.	OERX	Open Educational Resources
55.	PSDSARC	Prince Sultan Defense Studies and Research Center
56.	RDIA	Research, Development, and Innovation Authority
57.	RRA	Responsible Research Assessment
58.	SAIP	Saudi Authority for Intellectual Property
59.	SAMI	Saudi Arabian Military Industries
60.	SASO	Saudi Standards, Metrology, and Quality Organization
61.	Saudi NIH	Saudi National Institute of Health
62.	SDAIA	The Saudi Data and Artificial Intelligence Authority
63.	SEU	Saudi Electronic University
64.	SPInE	Strategic Partner in Innovation & Excellence
65.	STCEW	Saudi Telecommunication Company
66.	STI	Science, Technology, and Innovation
67.	SWA	Saudi Water Authority
68.	SWIC	Saudi Water Innovation Center
69.	TVTC	Technical and Vocational Training Corporation
70.	UOH	University Of Haeel
71.	UQU	Umm Al-Qura university
72.	WTIIRA	Water Technologies Innovation Institute and Research Advancement

A. GENERAL INFORMATION ABOUT THE RESPONDENT

- Country: The Kingdom of Saudi Arabia

- Organization responsible for the preparation of the report:

Name: King Abdulaziz City for Science and Technology (KACST) Website: [KACST](#)

Email address:

Phone number:

Please describe the role/mandate of your organization:

[Overview | KACST](#)

- Officially designated contact person who completed this survey:

Full name: Prof. Ahmed Mohammed Alabdulkader¹

Position: Advisor to the KACST President

Email address:

- Full Name of designated official certifying the report:

Project team

- Brief description of the consultation process established for the preparation of the report:

The established consultation process for the preparation of the national questionnaire on open science spanning over the period between August 2024 - February 2025, could be summarized as follows:

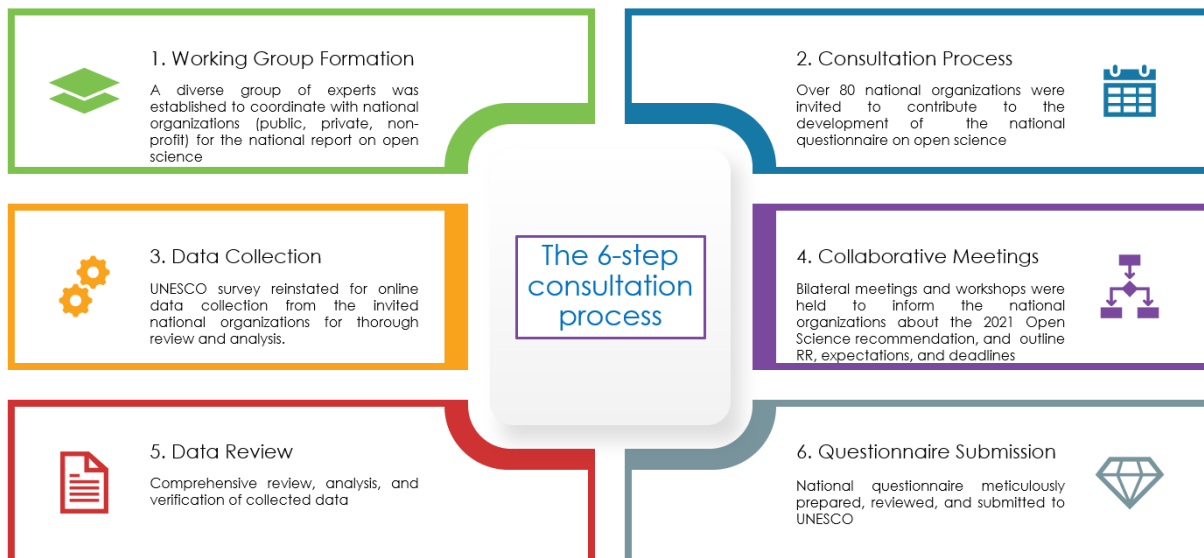
1. A working group consisting of a diverse array of experts and professionals was established to align and coordinate efforts with targeted national organizations, including public, private, and non-profit entities, to fulfill the requirements for preparing the national report on the implementation of the 2021 recommendation on open science at the national level.
2. Over eighty national organizations were invited to engage in the consultation process aimed at developing the national questionnaire on open science.
3. The UNESCO survey was reinstated to facilitate the online collection of data from participating national organizations, enabling a

¹ This survey was conducted by a team led by the officially designated contact person, as detailed on page 12 of this document.

thorough review and analysis of the gathered information.

4. A series of bilateral meetings and workshops were conducted to inform participating national organizations about the 2021 recommendation on Open Science – UNESCO, outlining roles, responsibilities, expectations, outputs, and deadlines.
5. The collected data underwent a comprehensive review, analysis, and verification process.
6. The national questionnaire was meticulously prepared, reviewed, and subsequently submitted to UNESCO.

The 6-step Consultation Processes for the Preparation of the National Questionnaire on Open Science – the Kingdom of Saudi Arabia



- Other organizations or entities including non-governmental organizations consulted to complete this survey:

Name of the organization: please see annex (1)

Website: please see annex (1)

Email address: please see annex (1)

Sector: (47) Public, (4) Private, (1) Civil Society, (4) Semi-government (3) non-profit organization.

The data presented in Figure (1) illustrates the distribution of invited national organizations participating in the national survey categorized by sector. Further insights are given to the distribution specifically within the governmental sector, as depicted in Figure (2).

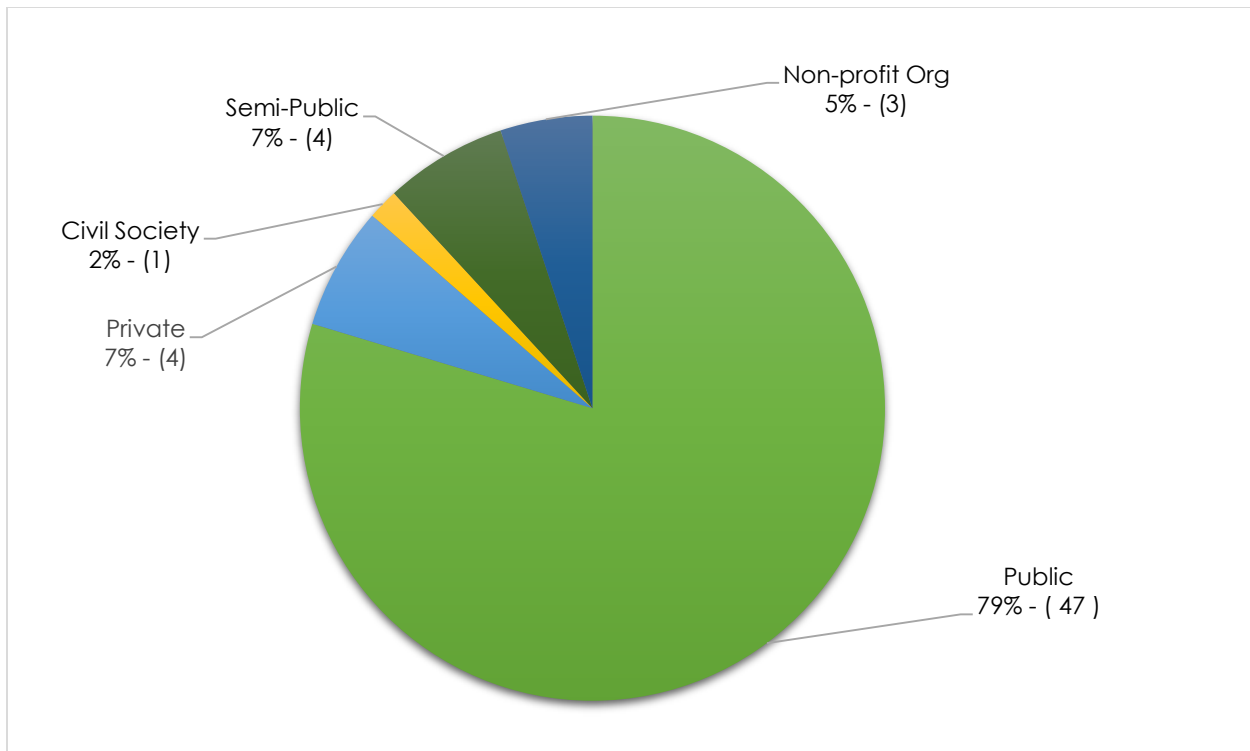


Figure (1): Distribution of the Consulted National Organizations in Saudi Arabia for Completing the 2021 UNESCO Recommendation on Open Science.

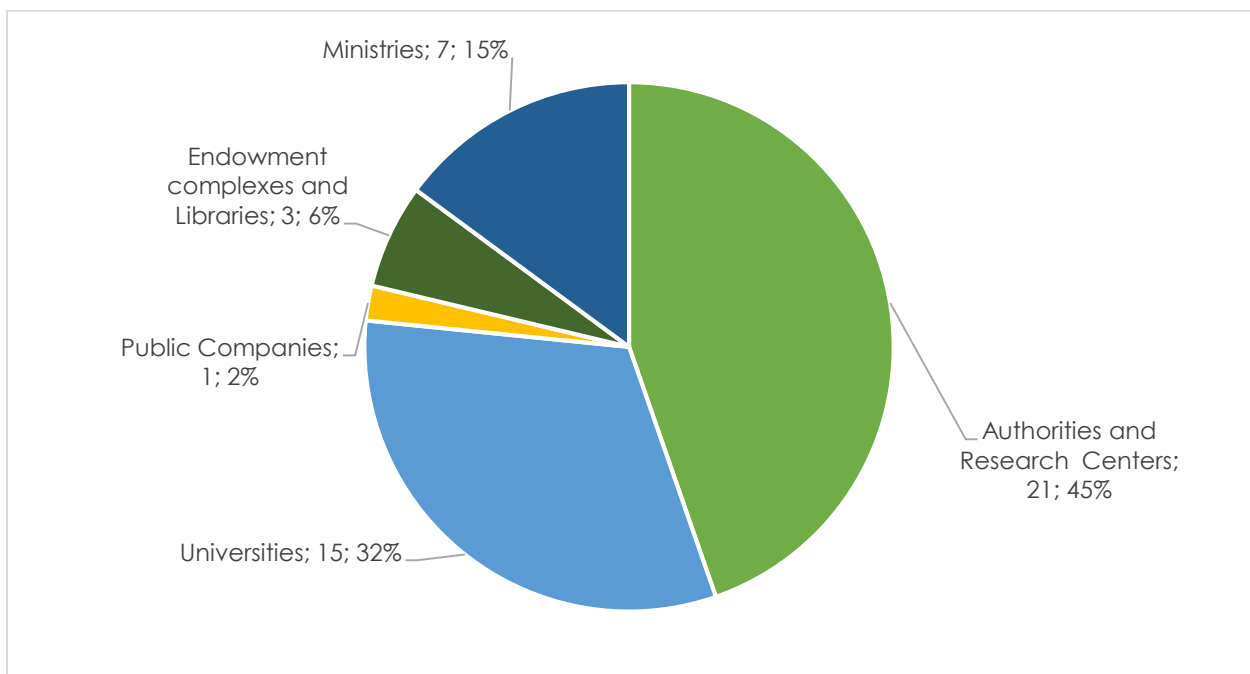


Figure (2): Distribution of the Consulted National Public Organizations in Saudi Arabia for Completing the 2021 UNESCO Recommendation on Open Science.

A.1 PROJECT TEAM

The 2021 UNESCO Survey on Open Science had been meticulously prepared, reviewed, and validated by the project subject matter expert team, led by the officially designated contact person, in collaboration with pertinent governmental organizations.

1. Prof Ahmad M. Alabdulkader
2. Prof. Hassan Y. Alayied
3. Mr. Abdullah S. Alkhaled
4. Dr. EMAN M. ALHUZIMI
5. Dr. Abdulaziz S. Alouffi
6. Dr. Ibraheem K. Hafiz
7. Ms. Atheer A. Alkubeyyer
8. Mr. Yazeed A. Alhajri

Supporting team:

1. Dr. Ahmed S. Haidyrah
2. Mr. Zaid M. Al-drees

A.2 ACKNOWLEDGMENTS

The project team extends its sincere gratitude to all national organizations involved in the completion of the UNESCO survey. A special acknowledgment is to Dr. Hamoud Alnughaymishi, Director of the Science Sector at the National Commission for Education, Culture, and Science, for his invaluable cooperation and support throughout the survey preparation process.

1. PROMOTING A COMMON UNDERSTANDING OF OPEN SCIENCE, ASSOCIATED BENEFITS, AND CHALLENGES, AS WELL AS DIVERSE PATHS TO OPEN SCIENCE.

1.1. Has the 2021 Recommendation been promoted and/or shared with appropriate ministries and institutions as well as affiliated organizations in your country?

Yes, Saudi Arabia, under the framework of Vision 2030, actively promotes open science as part of its broader national initiatives that prioritize research, development, and innovation (<https://vision2030.gov.sa/>). Several key institutions and ministries are instrumental in fostering open science and enhancing the collective understanding of its principles and advantages. Below is a list of the ministries and institutions dedicated to advancing the comprehension of open science:

- 1- King Abdulaziz City for Science and Technology (KACST) plays a vital role at the national level in promoting and sharing the 2021 UNESCO recommendation on Open Science. KACST coordinates efforts with various national organizations, including public, private, civil society, semi-government, and non-profit entities, to ensure the successful preparation of the national report on the implementation of this recommendation. This significant responsibility aligns with KACST's mandate as the officially designated national focal point for the preparation of the UNESCO 2021 recommendation on Open Science from August 2024 to February 2025, and a key national organization in enhancing the research, development, and innovation ecosystem within the Kingdom. Operating as a national laboratory and innovation park, KACST serves as the primary catalyst for the research and development sector and acts as a vital technical reference for both governmental bodies and the private sector. [Link](#).
- 2- The Ministry of Education (MOE) spearheads initiatives aimed at promoting open educational resources and plays a pivotal role in establishing the infrastructure essential for open science. This encompasses efforts to enhance data sharing via an open data platform, as well as fostering scientific literacy and advancing research development within the educational sector. [Link](#).
- 3- The Ministry of Communications and Information Technology (MCIT) plays a pivotal role in enhancing digital infrastructure and cultivating innovative

ecosystems that align with open science principles. Its efforts are directed towards establishing a thriving digital society by advancing skills and strengthening national digital capacities, thereby expediting the digital transformation process. These initiatives are integral to realizing the strategic objectives outlined in the Kingdom's Vision 2030. [Link](#).

- 4- The Research, Development, and Innovation Authority (RDIA) is committed to enhancing public access to research outputs, fostering the sharing of research data to improve collaboration and transparency, and facilitating partnerships among researchers, institutions, and industries on both national and international levels. [Link](#).
- 5- National universities and research institutions play a pivotal role in advancing open science practices by promoting open access to research and offering robust support for open data initiatives.
- 6- The National eLearning Center (NeLC) operates as an independent entity dedicated to promoting the values and principles outlined in the UNESCO Recommendation on Open Science. It actively collaborates with both public and private organizations to advance these ideals. NeLC facilitates its initiatives through a variety of channels, including webinars, workshops, and collaborative projects, while also maintaining an informative website. This strategy underscores NeLC's commitment to enhancing awareness and encouraging the adoption of open science principles across diverse sectors. [Link](#).
- 7- The Saudi Data and Artificial Intelligence Authority (SDAIA) is responsible for the development of the National Open Data Platform in Saudi Arabia. The primary objective of this initiative is to establish a public database that enhances transparency, promotes community engagement, and stimulates innovation. SDAIA collaborates with national partners to continually update the portal, striving to fulfill the goals outlined in the Open Data Strategy while supporting the overarching objectives of Saudi Vision 2030. This platform is designed to present datasets published by government ministries and agencies in an accessible and reusable format. [Link](#).

- 8- Community Engagement and E-participation: “The Government of Saudi Arabia acknowledges the significance of effective communication and constructive dialogue among the government, citizens, the business sector, and all other stakeholders as essential for accomplishing the goals of Vision 2030. Furthermore, the Saudi Government recognizes the value of digital interaction and participation as additional means of communication, enabling individuals to express their opinions, share ideas, and provide suggestions on specific societal topics. Therefore, the government values your input and wishes to encourage your active involvement. They rely on your creative ideas and helpful suggestions to enhance the quality of legislation, policies, strategies, initiatives, and services. Furthermore, Saudi Arabia acknowledges the crucial role of digital community engagement in improving government services that prioritize citizens' requirements. To harness the widespread telecommunications infrastructure and internet connectivity in Saudi Arabia, initiatives have been implemented to empower individuals to participate in crafting solutions to matters of public concern. The utilization of information and communication technology brings the advantage of enhanced community engagement in governance. It enables active involvement in tasks such as designing and implementing government services, shaping policies, and fostering inclusive decision-making processes. Public consultation platforms prioritize preservation and safeguarding of your personal information. We adhere to strict privacy and security measures, ensuring that you can confidently and securely express your perspectives and ideas without hesitation. Your views and opinions can be freely shared with ease, knowing that your information is protected and the process is reliable”. [Link](#).
- 9- The King Abdulaziz Complex for Endowment Libraries (KAWLA) is committed to executing a series of initiatives aimed at enhancing the reach of open science. This will be achieved through collaboration with the private sector, leveraging its expertise in this domain. Additionally, the Complex adopts a data analysis approach to assess beneficiary information derived from prior open science initiatives. This strategy aims to accurately identify the needs of the target audience, thereby fostering the engagement of diverse societal segments in the realm of open science. [Link](#).

1.2. Is the 2021 Recommendation on Open Science available in the national language(s) of your country?

Yes, the recommendation is available in nine languages, including Arabic. [Link](#).

1.3. Have there been awareness-raising activities on open science, including all its key elements, associated benefits, and challenges, organized or foreseen to be organized by the end of 2025 in your country by national authorities or entities?

Saudi Arabia has proactively undertaken and strategized numerous awareness-raising initiatives centered on open science, emphasizing its fundamental components, advantages, and challenges. These efforts, spearheaded by national authorities and associated organizations, are anticipated to persist and broaden by 2025. The actions implemented to integrate the values and principles of open science encompass, but are not limited to, the following:

1. Educational Platforms and Open Educational Resources (OERs)
 - a) Madrasati (affiliated with the Ministry of Education) provides diverse educational content for students across all levels ([Madrasati](#)). This platform indirectly helps raise awareness about open science principles, including transparency and teamwork. [Link](#).
 - b) OERx Platform ([OERx](#)): Supports the creation and development of open digital educational content through workshops and training programs.
2. Technical and Vocational Training Corporation (TVTC):
 - a) Conducts training courses and workshops on open science and intellectual property.
 - b) Implements workshops in partnership with the Research, Development, and Innovation Authority.
 - c) Launches research, development, and innovation centers within training institutions.
3. Saudi Data and AI Authority (SDAIA):
 - a) Organizes training programs and workshops on AI, data science, and open access.

- b) Hosts the National Programming and AI Olympiad (ATHKA) and International AI Olympiad (IAIO), promoting open collaboration in research.
- 4. National e-Learning Center (NeLC):
 - a) Shares research data via the National Open Data Platform.
 - b) Expands Open Educational Resources (OERs) through the OERx initiative.
 - c) Facilitates university integration through the ASAS Initiative, enhancing access to quality education.
 - d) Hosts Open Education Week annually in March ([NeLC](#)).
 - e) Collaborates with international organizations like OECD, World Bank, and UNESCO to promote inclusivity in research and education.
- 5. King Abdulaziz and His Companions Foundation for Giftedness and Creativity (Mawhiba): Provides open-access books, scientific resources, and articles on gifted education (Mawhiba).
- 6. King Salman Science Oasis (KSSO): Organizes activities such as the STEAM Festival, highlighting open science principles (KSSO).
- 7. Saudi Water Authority (SWA): Through the Water Technologies Innovation Institute & Research Advancements (WTIIRA)([Water Research Community](#)"), it promotes open science collaboration with universities, research institutes, and the private sector ([SWA](#)).
- 8. King Abdullah University of Science and Technology (KAUST):
 - a) 2014: Introduced an Open Access Policy.
 - b) 2015–2024: Hosts an annual Open Access Week.
 - c) 2017: Became the first Middle Eastern institution with an Open Access mandate and endorsed the OA2020 Initiative.
 - d) 2019: Organized the Open Science Conference and the First KAUST Library Saudi National Seminar on ORCID.
 - e) 2021–2024: Expanded Open Access publishing, collaborated with major publishers and launched the FAIRmat-KAUST Seminar Series.
 - f) 2024: Signed a landmark Open Access agreement with Wiley and enabled Open Access for AIP Publishing Journals.

9. King Khalid University (KKU):

- a) Online Open Courses – Free courses to enhance knowledge accessibility.
- b) E-Lab – Virtual scientific labs.
- c) KKU Shorts – Bite-sized educational content.
- d) KKU Skills – Skill-building programs.
- e) Open Discussions & Dialogues – Forums for stakeholders to advance Open Science.

10. Saudi Electronic University (SEU): Conducts workshops on open-access publication strategies and data-sharing practices (SEU Open Data).

11. King Abdullah Petroleum Studies and Research Center (KAPSARC):

- a) Publishes research in Arabic and English for free access.
- b) Shares research datasets (except proprietary data).
- c) Develops publicly accessible research models.
- d) Hosts multi-stakeholder workshops.

([KAPSARC Publications](#)).

1.4 Have specific actions been undertaken or are planned to be undertaken by the end of 2025 in your country to incorporate the values and principles of open science ii, in publicly funded research?

Saudi Arabia is proactively incorporating open science values and principles into publicly funded research through key government agencies and funding organizations, including but not limited to the followings:

1. Research, Development, and Innovation Authority (RDIA), the primary funding agency for research and innovation in Saudi Arabia, has established policies and frameworks to embed open science in government-funded research:
 - a) Open Access Policy for Government-Funded Research: Implemented a national policy requiring publicly funded research outputs to be open access.
 - b) Integration of Open Science in National Research Strategy: Embedding principles of equity, transparency, and accessibility in research funding guidelines.

- c) Public-Private Collaboration: Encouraging open science adoption through cross-sector partnerships.
 - d) Capacity Building: Conducting workshops and policy briefings to enhance open science compliance among funded researchers.
2. Saudi Data and Artificial Intelligence Authority (SDAIA) is embedding open science principles in AI research and data-sharing initiatives, including:
- a) Open Access Policies: Mandating open access to AI research publications and datasets funded by SDAIA projects.
 - b) Data Sharing Initiatives: Expanding the National Data Bank to enhance research access to government datasets.
 - c) International Collaboration: Partnering with UNESCO and IRCAI to align with global open science standards.
3. Ministry of Education: The National Center for Curriculum is working with the Ministry of Education on value-oriented projects and initiatives.
- a) Equity and fairness: various educational materials are provided for both genders, students with disabilities, and adults in continuing education

<https://moe.gov.sa/ar/mediacenter/Pages/ien.aspx>

<https://www.moe.gov.sa/ar/mediacenter/MOEnews/Pages/e-l-1443-987.aspx>

- b) Collective benefit: the Ain portal has been used to give all students access to digital content, and it contains open learning materials for all members of society.

<https://moe.gov.sa/ar/mediacenter/Pages/ien.aspx>

<https://www.moe.gov.sa/ar/mediacenter/MOEnews/Pages/e-l-1443-987.aspx>

- c) Equality of opportunities, equity, and fairness: the content is designed in a manner that is appropriate for different age groups. Also, there are representations of diverse populations of society and enhancing the national identity.

<https://moe.gov.sa/ar/mediacenter/Pages/ien.aspx>

<https://www.moe.gov.sa/ar/mediacenter/MOEnews/Pages/e-l-1443-987.aspx>

To incorporate the values and principles of open science, several initiatives have been launched. These actions focus on transparency, accessibility, and collaboration, aligning with open science principles. Below are key examples:

- a) The website of the Ministry of Education

<https://moe.gov.sa/ar/pages/default.aspx>

- b) Ein Portal: The Ein Portal provides free access to a wide range of digital educational resources for K-12 students and teachers. It also includes the Ein YouTube Channel, which features over 86,000 videos and has received more than 280 million views from users in over 30 countries, supporting global accessibility and knowledge dissemination.
<https://moe.gov.sa/ar/mediacenter/Pages/ien.aspx>
- c) The Virtual Kindergarten App: The Ministry of Education has launched a free electronic self-learning application called The Virtual Kindergarten, designed for children aged 3-6 years. The app's content was developed based on early developmental learning standards and includes 11 educational units, each containing a diverse series of learning elements. This application provides educational and developmental experiences for all children, whether registered or unregistered in kindergartens, both within and outside the Kingdom of Saudi Arabia.
<https://apps.apple.com/sa/app/%D8%B1%D9%88%D8%B6%D8%AA%D9%8A/id1560617527>
- d) The Ain Rawdaty Educational TV Channel: A specialized educational TV channel, IEN Rawdaty, has been launched specifically for the kindergarten stage, targeting children aged 3-6 years. The channel broadcasts engaging educational content tailored to the developmental characteristics of this age group, ensuring accessibility to education for all children anytime and anywhere. It serves children already enrolled in kindergarten as well as those awaiting admission, providing them with learning opportunities until they join formally. The channel is available to all children both within and outside the Kingdom of Saudi Arabia. <https://www.ientv.edu.sa/ar/rawdaty>
- e) Madares Platform: A platform that contains information about private schools in Saudi Arabia, enabling parents to choose a school based on certain criteria such as geographic location, tuition fees, financing solutions, and other features. <https://tetco.sa/ar/madares>
- f) Saudi Digital Library: The Saudi Digital Library provides open access to academic research and scholarly resources, ensuring that research papers and publications are accessible to all. This platform promotes open science by fostering transparency and enabling knowledge sharing across the scientific community.
<https://sdl.edu.sa/SDLPortal/Publishers.aspx>
- g) OERx by NELC: The OERx platform, developed by the National e-Learning Center (NELC), facilitates the sharing of Open Educational Resources (OERs). By offering teaching and learning materials freely, it creates a collaborative environment for educational institutions and contributes to open science and education.
<https://nelc.gov.sa/en/services/oerx/21>

- h) Open Data Platform: The Ministry of Education has launched an Open Data Platform that is publicly accessible, providing educational and research data to the public. This platform is part of a broader effort to promote transparency, innovation, and collaboration across the country. By making data available, it encourages research and academic collaboration, aligning with open science principles.
<https://open.data.gov.sa/en/home>
- 4. The National e-Learning Center (NeLC), serving as a funding entity for digital education and research initiatives, has integrated open science principles into various projects.
 - a) Expansion of OERx and ASAS Initiatives: Increasing the number of Open Educational Resources (OERs) and institutional partners.
 - b) Training Programs: Hosting webinars and workshops to guide researchers in open science practices.
 - c) Global Partnerships: Strengthening collaborations with OECD, World Bank, and UNESCO to bridge gaps between research and education.
 - 5. Technical and Vocational Training Corporation (TVTC) as a funding agency for applied research and training, has planned several 2025 initiatives:
 - a) Integration of Open Science in Public Research: Partnering with RDIA to ensure compliance with national open science mandates.
 - b) Workshops and Awareness Programs: Training institutions to adopt open research practices.
 - c) Innovation Centers: Establishing R&D hubs to foster open research, collaboration, and knowledge sharing.
 - 6. King Abdulaziz Complex for Endowment Libraries (KAWL) funds and supports open research projects focused on:
 - a) Open Access to Publicly Funded Research: Publishing research outputs with full open access.
 - b) Linguistic Diversity in Research: Ensuring publications are available in multiple languages.

7. NEOM Research and Innovation Program is developing shared research infrastructures for publicly funded projects, allowing universities, research institutions, and industry partners to collaborate on open science initiatives.
8. King Abdullah University of Science and Technology (KAUST) a major recipient of government research funding, is advancing open science through:
 - a) Open Access Mandates: Requiring funded researchers to publish in [open-access journals](#).
 - b) Data Sharing [Policies](#): Expanding FAIR data principles (Findable, Accessible, Interoperable, Reusable).
 - c) Upcoming 2025 Events:
 - 3-day workshop: "Uncovering Data Hidden Gems: Effective Research Data Management".
 - Webinar Series: "FAIR Research Data Management in Materials Science with <https://repository.kaust.edu.sa/home> FAIRmat and NOMAD".
9. King Khalid University (KKU) with funding aligned under RDIA's Open Science Policy, applies:
 - a) Mandating Open Access Compliance: Ensuring all publicly funded research adheres to national open-access requirements.
 - b) Data Repository Expansion: Encouraging researchers to store, share, and reuse research data.
 - c) Capacity Building: Hosting training sessions on research ethics, open data management, and intellectual property.

2. DEVELOPING AN ENABLING POLICY ENVIRONMENT FOR OPEN SCIENCE

2.1 Does your country have a national policy, strategy, or plan of action on science, technology, and innovation (STI)?

Saudi Arabia embarked on enormous national efforts supplemented with wide-spectrum global consultations to prepare a national research, development, and innovation strategy (RDI Strategy) that is aimed to fulfill the national aspirations by 2050, including:

- a. Become a top 26 nation and 1st in the MENA region by 2030 (according to the Global Innovation Index),
- b. Reach budget envelope of 115 B SAR in RDI expenditure by 2040 (equivalent to ~2.5% of GDP),
- c. have 140,000 researchers employed in KSA by 2040,
- d. Create ~360,000 jobs by 2050, (5) achieve ~230 B SAR GDP contribution by 2050.

The Research, Development, and Innovation Authority (RDIA) is responsible for implementing and overseeing the RDI strategy. The development process involved collaboration among various stakeholders, including government ministries, academic institutions, research centers, and industry partners. This collaborative approach ensured that the strategy addresses national priorities and leverages the Kingdom's strengths. The Strategy focuses on four national priorities:

- a. Health and Wellness: Advancing medical research and healthcare innovations to improve public health outcomes.
- b. Sustainable Environment and Supply of Essential Needs: Ensuring sustainable access to water, food, and energy resources.
- c. Energy and Industrial Leadership: Promoting advancements in energy technologies and enhancing industrial capabilities.
- d. Economies of the Future: Fostering innovation in emerging fields to diversify and strengthen the economy.

These priorities are designed to position Saudi Arabia as a global leader in research and innovation, contributing to economic diversification and addressing global challenges.

- a) The strategy emphasizes collaboration across government entities, academia, and industry to drive innovation and address the nation's strategic priorities.

- b) These initiatives reflect Saudi Arabia's commitment to fostering a robust STI ecosystem that supports sustainable development and economic growth.
- c) For more detailed information, please visit the official RDIA website: https://www.rdia.gov.sa/media/bf3h3psk/economies-of-the-future-rdi-national-priority-in-saudi-arabia-analytical-study_full-study.pdf

The RDI strategy, which is still under development and confidential at this stage, has replaced the previous National Plan for Science and Technology, which was approved and implemented in 2002 (att. 1).

The Digital Economy Strategy by the Ministry of Communication and Information Technology (MCIT) is another relevant strategy that supports the implementation of the RDI Strategy. Yet, it is still under accreditation.

Table (1) summarizes the Science, Technology, and Innovation (STI) Policies, Strategies, and Plans of Action at the National Level in Saudi Arabia.

Furthermore, Saudi Arabia represented by King Abdulaziz City for Science and Technology (KACST) participated in the development of several Statements of Principles (SOPS) and action plans of the [Global Research Council](#) (GRC) that addressed open science and some of its key elements, some of the relevant GRC SOPS and action plans are:

1. [Statement of Principles for Scientific Merit Review, 2012.](#)
2. [Action Plan Towards Open Access to Publications, 2013.](#)
3. [Review of Implementation of the Global Research Council Action Plan towards Open Access to Publications, 2014.](#)
4. [Statement of Principles on Peer/Merit Review, 2018.](#)
5. [Statement of Principles on Recognizing and Rewarding Researchers, 2023.](#)

Such Statements of Principles (SOPS) and action plans were endorsed and utilized as deemed necessary in the development or revision of the organizational practices or guidelines.

Table (1): Science, Technology, and Innovation (STI) Policies, Strategies, and Plans of Action at the National Level.

Title	Authority in Charge	Entities Involved	Key Areas/Sections	Link/attachment
National Research, Development, and Innovation Strategy	RDIA	All concerned national organizations and relevant stakeholders in national and international domains	The RDI Strategy aims to fulfill the national aspirations by 2050, including: a) Become a top 26 nation and 1st in the MENA region by 2030 (according to the Global Innovation Index), b) Reach budget envelope of 115 B SAR in RDI expenditure by 2040 (equivalent to ~2.5% of GDP), c) have 140,000 researchers employed in KSA by 2040, d) Create ~360,000 jobs by 2050, (5) achieve ~230 B SAR GDP contribution by 2050.	Under development
Digital Economy Strategy	MCIT	All concerned national organizations and relevant stakeholders in national and international domains	na	Under Accreditation

2.2 In your country, is there a policy, or a set of policies and/or legal frameworks at the national level that address open science or any of its key elements in line with the definition, values, and principles outlined in the 2021 Recommendation on Open Science?

Yes, a comprehensive array of policies and legal frameworks has been established at the national level to address open science and its fundamental components, in alignment with the definitions, values, and principles articulated in the 2021 Recommendation on Open Science. This includes, but is not limited to:

1. Open Access to Government-Funded Research Infrastructure in the Kingdom of Saudi Arabia by RDIA.
2. The National Intellectual Property Strategy by Saudi Authority for Intellectual Property (SAIP)
3. The [Data Governance Policies](#) adopted by The Ministry of Education.
4. The [e-learning strategy](#) and [OER Policy and Guidelines](#) developed by the National eLearning Center (NELC).

Some other policies and legal frameworks at the institutional level that include all relevant open science elements, namely, open access to scientific knowledge, open science infrastructure, open engagement of societal actors, and open dialogue with other knowledge systems, as follows:

1. Open Data Policy and Information Freedom, Universities' Collaboration Agreements and MOUs, Digital Transformation Strategy, and Energy

- Ecosystem RDI Governance and Roles by KACARE.
- 2. the RDI executive strategy plan by MEWA (Att. 2).
- 3. Innovation strategy and business model by Saudi Water Authority -Water Technologies Innovation Institute and Research Advancement (WTIIRA).
- 4. [ArabicThon](#) data sharing request and policy, ARAI Center strategy, and Research Publications Policy by KSGAAL
- 5. Open Data Policy by the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith.
- 6. The KAWLA work plan, and the Digital Access Policy for Manuscripts and Publications at the King Abdulaziz Complex for Endowment Libraries (KAWLA).
- 7. Internal Policy for Paper Publication Rewards by Elm Company.
- 8. Research and Innovation policy by NEOM.

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

On the other hand, national universities and research centers have developed a set of policies, strategies, and legal frameworks that are aligned with its academic and research priorities and support Saudi Vision 2030, as follow:

- 1. Science, Technology, and Innovation Strategy; and Research and Publication Policy at Al-Jouf University.
- 2. Thesis and dissertation archiving policy, open access policy for scholarly research publications, and research data management policy by KAUST.
- 3. Research Strategy by KFSHRC

Table (3) outlines policy and legal frameworks at the national university and research centers level that address open science or any of its key elements.

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
Open Access to Government-Funded Research Infrastructure	RDIA	RDIA coordinated with various stakeholders, including government agencies, research institutions, and universities	2023	na	Facilitate access to government-funded research infrastructure. Optimize the use of existing research facilities. Enhance the efficiency of government spending on research infrastructure. Maximize the return on investment in government-funded research infrastructure. Promote open innovation and reduce fragmentation within the research, development, and innovation (RDI) ecosystem. Attract top local and international researchers and scientists. Harmonize guidelines on open access to research infrastructure across the Kingdom	The policy addresses several key elements of Open Science, including: • Open Access: Facilitating access to research infrastructure and data. • Open Research Data: Encouraging proper data management and availability for review and reuse. • Open Collaboration: Promoting collaboration among researchers, both locally and internationally. • Open Innovation: Encouraging innovative practices through shared resources and knowledge.	link
National Intellectual Property Strategy	SAIP	All concerned national organizations and relevant stakeholders in national and international domains	2022	na	The National Intellectual Property Strategy outlined its plan and initiatives for the upcoming years to build an integrated intellectual property ecosystem that supports the economy based on innovation through four basic pillars, which are the generation of intellectual property, the management of intellectual property, the commercial investment of intellectual property, and the protection of intellectual property.	The National Intellectual Property Strategy (NIPST) contributes to open science by fostering a knowledge-based economy and promoting innovation through intellectual property (IP). Key ways in which this strategy supports open scientific collaboration include: • Encouraging Innovation & Knowledge Creation: ◦ Establishes an IP ecosystem that stimulates research, leading to the development of new	link

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
						technologies and industries. ○ Supports R&D initiatives that drive scientific discoveries and make them accessible for further advancements. ● Facilitating Data Sharing & Collaboration: ○ Aims to create a culture of respect for intellectual property, ensuring that scientific and technological knowledge can be shared responsibly. ○ Encourages universities, startups, and businesses to participate in IP-driven innovation, enabling a more open exchange of ideas. ● Building an IP Infrastructure for Open Science: ○ Focuses on IP administration and commercialization, ensuring that scientific research results are efficiently managed and utilized. ○ Promotes fast and high-quality registration of IP, making research outputs more accessible and usable. ● Increasing Public Engagement & Awareness:	

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
						○ Aims to raise awareness about intellectual property, ensuring that researchers and innovators understand their rights while promoting open access principles. ○ Supports educational initiatives to cultivate creative and competent talents, leading to a more engaged scientific community. ● Aligning with Global Best Practices: ○ Benchmarks against leading economies and integrates international best practices, ensuring that scientific and technological advancements align with global knowledge-sharing efforts. ○ Encourages cross-border IP cooperation, allowing researchers to collaborate internationally and leverage global scientific resources. By strengthening the intellectual property ecosystem, this strategy helps create an environment where scientific knowledge can be produced, shared, and utilized effectively, fostering global collaboration and accelerating technological progress.	

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
Data Governance Policy	MOE	Deputy for Planning and Development	Na	2021	• Data Classification Policy. • Personal Data Protection Policy. • Data Sharing Policy. • Freedom of Information Policy. • Open Data Policy. • Personal Data Protection Policy for Children and alike. • General Rules for Transferring Personal Data Outside the Geographical Borders of the Kingdom of Saudi Arabia	• Key principles of open data. • General rules of open data. • Roles and responsibilities. • Compliance	link
OER Policy and Implementation Guidelines	Nelc				• Accelerate the development and adoption of emerging technologies.	•	link
National eLearning Center Strategy					• Promote innovation and evidence-based research in digital learning. • Ensure quality and certification within the digital learning system. • Enhance international collaboration across various sectors.	•	link
Open Data Policy and Information Freedom	KACARE	KACARE and SDAIA	2022	2024	• Open access to scientific knowledge. • Open science infrastructure.		Internal website
Universities' Collaboration Agreements and MOUs		KACARE and Universities (KSU, KFUPM, KAU, PNU)	2020	2024	• Open engagement of societal actors. • Open dialogue with other knowledge systems		Restricted
Digital Transformation Strategy		KACARE	2024	2024			Restricted

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
Energy Ecosystem Governance and Roles RDI	MOENERGY - KACARE		2023	2024	See diagram	Aligns with the principle of open access (16.a, 16.f, 16h).	
The Executive Strategy Plan RDI	MEWA	The Strategic Plan was guided by 150+ stakeholder engagements, including Ministers, Vice Ministers, RDI Committee Members, a Diverse panel of experts and advisors, and Wider ecosystem senior leadership.	2023		The strategy plan aims to propel research and innovation in the environment, water, and agriculture sectors by steering the ecosystem and through targeted instruments that enhance linkages, support the deployment of technologies, and increase the responsiveness of demand. It contains the main elements of open science: • Open access to scientific knowledge. • Open engagement of societal actors. • Open dialogue with other knowledge systems		Att. 2
Innovation strategy and business model	WTIIRA				The strategy and business model support science (innovation). It comprises goals - one water, circular pathways (sustainable water), and building an ecosystem spanning 3 - 5 years.		
ARAI Center strategy	KSGAAL				focuses on supporting researchers in Arabic natural language processing.		Under accreditation
Research Publications Policy	KSGAAL						Link

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
ArabicThon data sharing request and policy	KSGAAL				Request any applications developed for the hackathon to adhere to open-source principles.		Link
Open Data Policy	HADITH	The Permanent Steering Committee for Data Management and Governance	2024	2024	The policy aims to: - Enhancing transparency between the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith and beneficiaries. - Enabling the creation of added value from data. - Empowering individuals to gain a better understanding of how the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith operates by providing access to open data. - Encouraging innovation and economic growth by enabling institutions and individuals to utilize data to develop new value-added services. - Stimulating and supporting scientific research through conducting studies and research based on published information. Accordingly, the policy contains principles of open	General Rules During the Open Data Lifecycle Stages	

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
					data policy, roles, and responsibilities.		
Work plan	KAWLA	The General Department of Information Technology in cooperation with the General Department of Research and Studies			The work plan aims to provide open scientific knowledge, in addition to several forms of infrastructure for open science in cooperation with many relevant parties. The work plan includes all relevant open science elements, namely, open access to scientific knowledge, open engagement of societal actors, and open dialogue with other knowledge systems.		UD ²
Digital Access Policy for Manuscripts and Publications	KAWLA				Digital Access Policy for Manuscripts and Publications at the King Abdulaziz Complex for Endowment Libraries approved by His Royal Highness the Chairman of the Board of Trustees of the Complex in letter No. (1357), dated (2/2/1444 AH), and includes the provision of supporting models from the information containers owned by the Complex through its digital repository.		I DAR
Internal Policy for Paper Publication Rewards	Elm Company	The policy was developed by the Research Committee with input from faculty members and	2022	2023	Promotion of Open Access Publishing: Encouraging company researchers to publish findings in open-access journals or platforms to	Open Access to Scientific Knowledge: The policy promotes open access by encouraging publications in open-access journals and platforms.	

² UD = Under Development

Table (2) summarizes the policy and legal frameworks at the national or institutional levels that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/ Sections	Elements of Open Science Addressed	Link
		administrative staff. The development process included consultations, reviews, and alignment with the organization's strategic goals to promote research excellence.			enhance visibility and contribute to the scientific community. • Incentive Structure: Providing rewards and recognition for employees who publish in reputable and open-access venues. • Eligibility Criteria: Defining the types of research outputs that qualify for rewards, including publications, patents, and conference presentations. 4. Application and Evaluation Process: Outlining procedures for submitting publications for consideration and criteria for evaluating the impact and quality of the research.	• Open Engagement of Societal Actors: Not fully addressed but recognized as an area for future development.	
Research and Innovation Policy	NEOM					The policy includes all relevant open science elements, including open access to scientific knowledge, open science infrastructure, open engagement of societal actors, and open dialogue with other knowledge systems.	UD ³

³ UD = Under Development

Table (3) Policy and legal frameworks at the National University and Research Centers Level that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/Sections	Elements of Open Access	Link
Science, Technology, and Innovation Strategy	JU	Internal stakeholders: Academic faculties, research centers, and administrative departments. - External stakeholders: Ministry of Education, King Abdulaziz City for Science and Technology (KACST), and industry partners.	2020	2024	- Promoting high-quality research and innovation. - Encouraging open access to scientific knowledge. - Strengthening open science infrastructure. - Fostering collaboration with societal and industry actors. - Supporting the development of a knowledge-based economy		
Research and Publication Policy	JU	Internal stakeholders (research departments, academic councils) in alignment with national guidelines from the Ministry of Education and Vision 2030, and consultation with faculty, researchers, and administrative leaders.			- Open Access (16.a): Encourages researchers to publish in open-access journals and make their work available in institutional repositories. - Open Science Infrastructure (16.h): Development of digital repositories and platforms for storing and sharing research data. - Multilingualism (16.d): Policies promote research outputs in Arabic and English to increase accessibility.	Open Access to Scientific Knowledge Open Science Infrastructure. Al-Jouf University has a plan to: - Development of a formal Open Science Policy by 2025, aligned with UNESCO's recommendation. - Expanding initiatives to include citizen science projects and stronger partnerships with the private sector and indigenous knowledge holders. - Increasing funding and infrastructure for open science practices, including multilingual dissemination and stakeholder engagement. Al-Jouf University remains committed to advancing its open science practices and aligning them with international standards to foster a culture of transparency, collaboration, and inclusivity in scientific research.	Link

Table (3) Policy and legal frameworks at the National University and Research Centers Level that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/Sections	Elements of Open Access	Link
KAUST Thesis and Dissertation Archiving Policy	KAUST	Library, Registrar, Academic division	2010		• Open access to scientific knowledge. • Open science infrastructure:		link
Open Access Policy for Scholarly Research Publications	KAUST	Library, Registrar, Academic Division.	2014		• Open engagement of societal actors: • KAUST is a member of The Coalition of Networked Information (CNI) • Scholarly Publishing & Academic Resources Coalition (SPARC) • Research Data Alliance (RDA) • Open Researcher and Contributor Identifier (ORCID) • International Council on Archives (ICA) • DRYAD - an international open-access repository of research data • DataCite - an international organization that supports data citation • Digital Curation Centre (DCC) • Open Education Consortium • NOMAD, Materials science data managed and shared • FAIRmat, open data for condensed-matter and chemical physics of solids	The KAUST Open Access Publishing Program supports its researchers by establishing a range of OA agreements with publishers to cover or discount article processing charges (APCs) to ensure our research articles on world-leading scientific advances are available to all. KAUST is partnered with MDPI to provide discounts to APCs for researchers publishing in MDPI journals. MDPI's Institutional Open Access Program (IOAP) offers flexible agreement models for institutions, making it easy for authors to publish in CC-BY OA journals.	
KAUST Research Data Management policy	KAUST	Offices of the President, the Provost, the Vice-President for Research, Deans of Divisions, Library, IT, Super Computing Labs, Research Compliance, Core Labs, Research Centers	2025				

Table (3) Policy and legal frameworks at the National University and Research Centers Level that address open science or any of its key elements.

Title	Authority in Charge	Entities Involved	Year of Adoption	Year of Last Update	Key Areas/Sections	Elements of Open Access	Link
					• FAIR-DL, Data Infrastructure for Physics, Chemistry, Materials, and Astronomy • The Declaration on Research Assessment (DORA) • Open Access 2020 Initiative open dialogue with other knowledge systems		
Research Strategy	KFSHRC				• Open access to scientific knowledge. • Open engagement of societal actors. • Open dialogue with other knowledge systems.		UD

2.3 In your country, are there specific policy instruments that aim to promote open science?

Yes, several policy instruments have been developed or underway to promote open science at the organizational level, including but not limited to the following:

1. 2025 Research Grants on Research, Development, and Innovation (RDI) National Priorities, by RDIA.
2. Open data policy by SIAP.
3. Open data platform by MOE.
4. Open data privacy Policies by MOE.
5. Energy Research, Innovation, and Development digital platform by the Ministry of Energy (MOENERGY).
6. University Collaboration Program by KACARE.
7. Technology Localization Initiative by KACARE.
8. Renewable Energy Resources (Atlas) Portal and Physical IOT Devices by KACARE.
9. Periodic Energy Observatory News by KACARE.
10. The Sectorial Human Capacity Development Portal by KACARE.
11. Renewable Energy Business Incubator by KACARE.
12. Renewable Energy Hackathon by KACARE.
13. Innovation Driven Water Sustainability by WTIIRA.
14. Water Research Community by WTIIRA.
15. Open-access platform by IDS.
16. Open Educational Resources Platform (OERx) by NeLC
17. [Licensing Regulations for eLearning and Training Programs](#) by NeLC
18. Open data policy by HADITH.
19. Open national center providing external and Internal service by King Fahad National Library (KFNL).
20. Open data-sharing frameworks and platforms by SDAIA.
21. Integration with National Initiatives by TVTC.
22. Supporting Research and Development by TVTC.
23. Enhancing Open Educational Resources by TVTC.
24. Awareness and Training by TVTC.
25. Strategic Partnerships by TVTC.

Table (4) summarizes specific policy instruments that aim to promote open sciences fully or partially in Saudi Arabia.

On the other hand, some national universities and research centers developed specific policy instruments that aim to promote open sciences fully or partially in Saudi Arabia.

1. E-Lab (Electronic Laboratory Platform) by King Khalid University (KKU)

2. Open Courses for All (OCW) by King Khalid University (KKU).
3. Shorts Courses by King Khalid University (KKU).
4. Skills Platform (UpSkill) by King Khalid University (KKU).
5. Open Science Strategy and Policy Framework by Al-Jouf University (JU)
6. Publishing and Deamination of large Research dataset by KAUST.
7. Developing Material Science Research Data Repository by KAUST.
8. General guidelines and 'best practices' by KAPSARK.

Table (5) summarizes the specific policy instruments that aim to promote open sciences fully or partially at the national universities and research centers in Saudi Arabia.

Table (4) Specific Policy Instruments that aim to promote open sciences fully or partially in Saudi Arabia.

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
Research Grants on Research, Development, and Innovation (RDI) National Priorities	RDIA	support pioneering research and innovations addressing national priorities, thereby strengthening Saudi Arabia's leadership in the RDI sector	The grant programs encourage collaboration between academia, industry, and government entities, fostering open innovation and the dissemination of research outcomes.	na
Open data policy	SAIP	This policy aims to achieve the principle of transparency and enrich the community by providing access to datasets related to intellectual property.	• Transparency & Accessibility: The policy promotes public access to intellectual property-related datasets, ensuring transparency and enriching the community with open scientific knowledge. • Data Sharing & Collaboration: It commits to making datasets freely available and allows individuals and organizations to request additional data, fostering collaboration across scientific and innovation ecosystems. • Responsible Data Use: The policy ensures the ethical use of data, prohibiting political, criminal, or discriminatory applications, aligning with responsible science. • Data Integrity & Attribution: Requires users to credit the source and maintain data quality, supporting a focus on scientific integrity and trustworthiness. • Adaptability & Continuous Review: The policy includes regular updates to keep up with best practices, ensuring compliance with	Att. 3

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
			evolving global open science standards. • National & International Alignment: It follows Saudi Arabia's national open data policies, which align with broader global frameworks, including Open Science recommendations. By facilitating free, ethical, and high-quality data sharing, this policy contributes to the global open science movement, promoting equity in access to knowledge and fostering international scientific cooperation.	
Open Data Platform	MOE	This platform aims to promote transparency, innovation, and collaboration, encouraging academic research and partnerships in line with open science principles.	This platform aims to promote transparency, innovation, and collaboration, encouraging academic research and partnerships in line with open science principles.	Link
Open Data Privacy Policy	MOE	The policy aims to protect sensitive information, the Ministry has implemented data privacy policies. These policies ensure that while confidential educational data is safeguarded, non-sensitive information can still be shared for research, innovation, and educational growth, supporting a balance between openness and security.		Link
Energy Research, Innovation, and Development of digital platform	MOENEREGY	Encourage knowledge sharing between energy stakeholders both from the public and private sectors and foster collaboration between entities on advancing innovative solutions to address current challenges and open new opportunities in the energy sector.	Scientific knowledge sharing, open infrastructure usage, and collaboration with public and private organizations.	
Universities Collaboration Program	KACARE	The objective of the program is to promote collaboration work with universities on innovation research, Intellectual Properties, and open use for research and technical infrastructures and tools.	Scientific knowledge sharing, open-access publications, and open infrastructure usage	
Technology Localization Initiative	KACARE	The objective of the program is to work with public and private entities on technology development projects in renewable energy.	Scientific knowledge sharing, open access publications	
Renewable Energy Resources (Atlas) Portal	KACARE	The objective of the program is to promote Renewable Energy technologies and localization development and projects.	Scientific knowledge sharing, open-access publications, open infrastructure usage, and	

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
and Physical IOT devices			collaboration with public and private organizations	
Periodic Energy Observatory News	KACARE	The objective of the publication is to promote Renewable and Atomic Energy technologies and support Knowledge exchange, by offering cutting-edge technologies and new findings news on the world.	Scientific knowledge sharing and open-access publications	
The Sectorial Human Capacity Development Portal	KACARE	The objective of the portal is to promote and enrich the energy field environment in the Kingdom via the portal's content and services.	The portal will offer scientific knowledge sharing, open educational resources and learning, and skill development capabilities. Additionally, it will support an international energy community of experts, researchers, and individuals.	UD
Renewable Energy Business Incubator	KACARE	The objective of the business incubators is to support innovation and entrepreneurship in renewable and atomic energy by fostering SMEs, localizing expertise, enhancing intellectual capital, and increasing local content contributions in these fields.	The incubators will promote knowledge sharing; provide open resources, and offer training and mentorship to develop national skills. They will also connect innovators with the global energy community to drive collaboration and innovation.	
Renewable Energy Hackathon	KACARE	The primary objective is to bring together bright minds, innovators, and industry experts to collaboratively develop solutions that address challenges and accelerate the adoption of renewable energy technologies in Saudi Arabia.	The Renewable Energy Hackathons will promote scientific knowledge sharing, provide open educational resources, and focus on skill development through workshops, training, and mentorship. It will also connect participants with the international energy community, fostering collaboration and innovation in renewable energy.	
Innovation-Driven Water Sustainability	WTIIRA	It is an annual event that aims to gather local and international universities to share their ideas and discuss challenges and opportunities for collaboration.		
Water Research Community	WTIIRA			
Open-access platform	IDS	It is an open-access platform for all IDS's publications, journals, and reports.		link link link
Open Educational Resources Platform (OERx)	National eLearning Center	To provide accessible, high-quality educational materials to learners, educators, and institutions, fostering a culture of open knowledge sharing and collaboration.	Open Access to Scientific Knowledge: The platform offers free access to a wide range of educational resources, aligning with open science principles.	OERx Platform

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
			- o Open Science Infrastructure: OERx serves as a digital repository, facilitating the storage, management, and dissemination of educational content. - o Open Engagement of Societal Actors: The platform encourages contributions from educators and institutions, promoting collaborative development and sharing of resources.	
Licensing Regulations for eLearning and Training Programs	National eLearning Center	To establish standards and guidelines for the development, delivery, and quality assurance of eLearning and training programs, ensuring they meet national and international benchmarks.	- o Open Access to Scientific Knowledge: By setting quality standards, these regulations ensure that eLearning programs provide reliable and accessible scientific knowledge. - o Open Science Infrastructure: The regulations support the development of robust eLearning infrastructures that facilitate open access and sharing of educational content.	link
Open Data Policy	HADITH	• Enhancing transparency between the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith and beneficiaries. • Enabling the creation of added value from data. • Empowering individuals to gain a better understanding of how the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith operates by providing access to open data. • Encouraging innovation and economic growth by enabling institutions and individuals to utilize data to develop new value-added services. Stimulating and supporting scientific research through conducting studies and research based on published information.	The Principles of Open Data Policy, Roles and Responsibilities. General Rules During the Open Data Lifecycle Stages	
open national center	KFNL	Enhancing and making information resources available to all researchers in different ways: 1. External service:		

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
		- KFNL supports open access that allows researchers access from outside the library through the library's website and provides searching techniques with the highest quality to access knowledge resources. https://kfnl.gov.sa/ar/Pages/default.aspx 2. Internal service: - KFNL allows access inside the library through the digital repository to view all the digital content for different information resources. https://dlibrary.kfnl.gov.sa/home/?locale=ar - Fair use under Copyright Laws such as Saudi Copyright Protection law under article 15. https://laws.boe.gov.sa/BoeLaws/Laws/LawDetails/67d159e6-ee98-4efc-a2ee-a9a700f17083/1, and according to International conventions such as the Berne Convention. https://www.wipo.int/wipolex/en/treaties/textdetails/12214		
Open data-sharing frameworks and platforms	SDAIA	The policy instruments aim to promote open science, including initiatives to establish open data-sharing frameworks and platforms, with a focus on enhancing transparency, accessibility, and collaboration in AI and data science research.	The policies are being designed in alignment with national guidelines and international recommendations on open science, with an emphasis on supporting FAIR (Findable, Accessible, Interoperable, and Reusable) data principles.	UD ⁴
Integration with National Initiatives	TVTC	Collaborating with the Research, Development, and Innovation Authority (RDIA) to develop policies that support open access to publicly funded research outcomes	Policies to promote open science in alignment with national directions and Saudi Vision 2030.	UD
Supporting Research and Development		Establishing research centers and training institutions aimed at enhancing transparency and collaboration in research.		UD
Enhancing Open Educational Resources		Proposing policies to support the development and use of open educational resources at the national level		UD
Awareness and Training		Offering training programs and workshops to raise awareness of open science principles and practices among researchers and trainees.		UD

⁴ UD = Under Development

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
Strategic Partnerships		Strengthening collaboration with national and international institutions that have successfully adopted open science practices.		UD

Table (5) Summarizes the Specific Policy Instruments that aim to promote open sciences fully or partially at the National Universities and Research centers in Saudi Arabia.

Title	Authority in Charge	Objectives	Elements of Open Science Addressed	Link
E-Lab (Electronic Laboratory Platform)	KKU	The objective of the E-Lab: • Provide virtual labs for students and researchers to conduct experiments and access practical resources. • Eliminate the constraints of physical access to labs	These platforms represent KKU's commitment to fostering open science principles, expanding access to knowledge, and enhancing global collaboration. These include open access to educational content and research community engagement in learning and teaching, and promoting equitable access to scientific and educational resources.	link
Open Courses for All (OCW)	KKU	The objective of the OCW: • Deliver free, high-quality educational courses and materials, enabling individuals to learn and study independently, regardless of their affiliation with the university.	• Open access to educational content and research. • Community engagement in learning and teaching. • Promoting equitable access to scientific and educational resources.	link
Shorts Courses	KKU	To provide concise, topic-focused videos and materials for quick learning and skill-building in various scientific and educational domains.	• Open access to educational content and research. • Community engagement in learning and teaching. • Promoting equitable access to scientific and educational resources. •	link
Skills Platform (UpSkill)	KKU	• To deliver specialized skill-based training and capacity-building courses, focusing on enhancing students' and researchers' abilities in technical and professional fields	•	link
Open Science Strategy and Policy Framework	JU	The objective is to: • Establish a structured approach for implementing open science principles across all research activities. • Promoting access to scientific knowledge. • Fostering collaboration with individuals.	• Open Access: Encouraging and supporting researchers to publish in open-access journals. • Infrastructure Development: Creating institutional repositories for storing and sharing research data. • Stakeholder Engagement: Establishing partnerships with individuals and private sectors to implement open science practices. • Multilingualism: Promoting research dissemination in both Arabic and English to ensure inclusivity. The planned actions are to: • Finalizing the Open Science Strategy by 2025, including clear indicators, short-term deliverables, medium-term results, and long-term impacts. • Integrating open science practices into the university's research policies and operations. • Launching workshops and training programs to raise awareness and equip researchers with the necessary tools and knowledge. Al-Jouf University remains committed to fostering a legal and operational environment that supports open science principles while addressing existing challenges. The finalized policy instruments will align with Saudi Vision 2030 and international standards for open science.	na

Publishing and Deamination of large Research dataset	KAUST	Enabling KAUST Researcher to share large volumes of research datasets.	Records were created at the institutional repository with descriptive metadata, DOI minting for each dataset supplemental files, and discoverable metadata.	
Developing Material Science Research Data Repository	KAUST	A platform dedicated to managing, sharing, and publishing materials and science research data	It offers a suite of tools and services designed to facilitate data handling and collaboration within the materials science community.	
General guidelines and 'best practices'	KAPSARC		Section 5: Publishing/General guidelines "h) Open Access: In general, all KAPSARC research publications will be made publicly accessible and free to reuse under the sole condition that the source is properly cited. This applies to: KAPSARC publications including research articles authored or co-authored by KAPSARC personnel that are published in external Peer Reviewed literature. Authors of articles or other publications such as books, being published by external parties, should connect with the KAPSARC K&A Library to organize Open Access arrangements, publishing permissions on the KAPSARC website, and any other required licenses. In the case of models, data, and tools, the SP team shall publish following the Intellectual Property guidelines of KAPSARC." Section 5.2: Publishing a research paper externally (page 25) "c) The K&A Library is responsible for arranging open access for papers published externally."	

2.4 In your country, is there a national implementation plan, strategy, policy, or roadmap for the implementation of open science at the national level in line with the 2021 Recommendation?

Yes, there are considerable efforts at the national and institutional levels that promote the implementation of open science. Such efforts could be summarized as follows.

1. RDIA is actively developing and implementing strategies to promote Open Science in Saudi Arabia, aligning with the 2021 UNESCO Recommendation on Open Science. The National Research, Development, and Innovation (RDI) Strategy aims to position Saudi Arabia as a global leader in research, development, and innovation by fostering a knowledge-based economy. The Strategy contains the key elements of open science, namely:
 - a. Open Access: Promoting the availability of research outputs to the public.
 - b. Open Data: Encouraging the sharing of research data to enhance collaboration and transparency.
 - c. Open Collaboration: Facilitating partnerships among researchers, institutions, and industries both nationally and internationally. For more detailed information, you can refer to the RDIA's publication: <https://www.rdia.gov.sa/media/hn2boqdr/insights-into-saudi-arabia-s-research-and-innovation-ecosystem.pdf>

This document provides comprehensive insights into the Kingdom's strategies and initiatives in advancing research, development, and innovation.

2. The Saudi Data and Artificial Intelligence Authority (SDAIA) has developed the National Strategy for Data and Artificial Intelligence (NSDAI), which outlines a comprehensive roadmap to position Saudi Arabia as a global leader in data-driven economies. This strategy emphasizes the development of open data policies, enhancement of data infrastructure, and promotion of data sharing and transparency, aligning with the principles of open science. [here](#)
Additionally, SDAIA has established the National Data Management Office (NDMO) to oversee data governance and implement policies that support open data initiatives. The NDMO's efforts include developing standards and regulations to facilitate data sharing and accessibility, further supporting the implementation of open science practices. [here](#), and the "National Data Index (NDI)" aimed at monitoring and evaluating the maturity of data management practices in government entities and their compliance with national data management controls and operational indicators [here](#).

3. The strategic plan of King Abdulaziz and his Companions Foundation for Giftedness and Creativity (Mawhiba), emphasizes providing its educational services and scientific resources free of charge to our targeted audience across the nation and the entire Arab region, regardless of gender.
4. Saudi Water Authority (SWA) within the limits of legal framework and contractual obligations implements and will continue to implement its mission for open science adoption. Water Technologies Innovation Institute and Research Advancement (WTIIRA) is a research and development arm of SWA initiating and implementing open science (<https://www.swa.gov.sa/en/laborateis-watiira>).
5. WTIIRA has a 3 to 5-year strategy that includes a building ecosystem goal under the principles of open science (innovation). It is now developing a high-level operating model and roadmap.
6. KACARE has been mandated (Items: 1) to participate in the development of national plans and necessary strategic implementation plans for the peaceful use of Atomic Energy ("AE") and Renewable Energy ("RE") technologies and applications. Along with, proposing regulations and policies related to KACARE's role. The KACARE's operational plan and the executive plan for the open Innovation was developed in cooperation with the Ministry of Energy, Ministry of Education, Ministry of Human Resources and Social Development, RDIA, and KACST, universities and Institutes, SIRA, SEEC, HRDF.
7. The NEOM regulatory framework is presently under development and will include a slate of legal provisions that will impact open science. Pursuant to the Research and Innovation policy framework, the R&I strategy and operational plan will subsequently be developed and implemented as NEOM U and the other pillars of the ERI sector are activated, and gain in maturity and scale.
8. LCGPA: Strategies, policies, and plans have been developed for science, technology, and innovation, and were formulated through a series of workshops and consultations with key stakeholders, ensuring a comprehensive approach that incorporates diverse perspectives and expertise.
9. The NeLC has an implementation strategy that aligns with the 2021 UNESCO Recommendation on Open Science. The [Digital Learning Strategy](#) includes a core strategic pillar focused on Collaboration, which emphasizes encouraging coordination and resource sharing to support the digital learning ecosystem. This is achieved through objectives such as

collaborative sharing of data, content, capabilities, and expertise, and strengthening regional and global impact. Building on this, the [National eLearning Center Strategy](#) which is directly based on the Digital Learning Strategy, features a key strategic pillar pertaining collaborative ecosystem, which promotes intersectional and international collaboration. Coordination in the development, curation, and sharing of curriculum and capabilities. These two approved strategies, which guide our efforts for the next four years, provide a clear roadmap for fostering collaboration and resource sharing in alignment with the principles of open science. They ensure that our national and organizational efforts are geared toward transparency, inclusivity, and collective benefit in education and research.

10. The strategy of the Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith (HADITH) includes initiatives and projects aimed at disseminating scientific outputs and making them accessible to beneficiaries.
11. The KSSO's primary mission is to raise public awareness about science and technology. Future plans include enhancing collaboration with policy-making organizations to align with national open science objectives.

Furthermore, there are some efforts from the national universities and research centers to develop implementation plans, strategies, policies, or roadmaps for the implementation of open science at the national or organizational level in line with the 2021 Recommendation, which could be highlighted as follows:

1. Al-Jouf University is currently in the process of developing an implementation plan and roadmap for open science in line with the 2021 UNESCO Recommendation on Open Science. The roadmap aims to incorporate open science values and principles into the university's research ecosystem, ensuring equitable sharing of knowledge and addressing challenges such as data exploitation, privacy, and inequalities in research funding. The Council of Universities Affairs, and Research, Development, and Innovation Authority (RDIA), other universities, industry partners, and community organizations are among the supporting national entities to the implementation plan.
2. KAUST's New Strategy has been developed to achieve stronger international collaboration & partnerships, global integration and open access policies, collaboration with high-tech, high-talent areas through

hubs and partnerships, and platforms to accelerate innovation using world-leading research standards. KAUST supports research publishing by offering many open-access publishing agreements in 2024. These transformative agreements allow KAUST's corresponding authors to publish their accepted articles in open-access mode and thus be made openly available at no cost to the author or with a discount. The library subscription covers the publishing charges in "Read and Publish" deals. APC (article processing charges) discount deals provide an agreed discount on the APC, while KAUST authors pay the discounted APC. When KAUST doesn't have an open-access publishing deal with the publisher, KAUST authors are advised to submit an author-accepted version of their paper to KAUST Research Repository. KAUST Library follows the recent trends in open-access movements. It has established many read-and-publish agreements to manage hybrid journal publishing and institutional agreements to support gold-only publishers.

2.5 In your country, are there policies and/or strategies that promote open science in line with the 2021 Recommendation at the institutional level, including in the context of research-performing institutions, universities, scientific unions and associations, and learned societies?

Yes, Saudi Arabia adopted several policies and strategies that promote open science at the institutional level, in the context of research-performing institutions, universities, scientific unions, associations, and learned societies, including the following:

1. RDIA sets out national principles on [open access to government-funded research infrastructure](#) in the Kingdom of Saudi Arabia. These national principles aim to set out national principles to be used as a reference by all host organizations managing government-funded research infrastructures in the Kingdom of Saudi Arabia and external users seeking access. It also aims to help achieve the following objectives:
 - a) Facilitate RDI players' access to government-funded research infrastructure.
 - b) Optimize the use of existing government-funded research infrastructure.
 - c) Enhance the efficiency of government spending on research infrastructure.
 - d) Maximize return on investment in government-funded research infrastructure
 - e) Promote open innovation, reducing RDI ecosystem fragmentation.
 - f) Attract top local and international researchers and scientists.

- g) Harmonize guidelines on open access to government-funded research infrastructure across the Kingdom.
2. KACST sets several directions that address open science, including but not limited to:
 - a) The open access policy for KACST RD infrastructure.
 - b) Governing regulation to host companies.
 - c) Open data management and classification policies.
 3. SDAIA has implemented several policies and strategies that promote open science in line with the 2021 Recommendation. These initiatives include:
 - a) Open Data Policy: SDAIA has established an Open Data Policy that sets the regulatory framework for open data, a subset of public information. This policy aims to increase transparency, promote innovation, and support the realization of Vision 2030's goals by enhancing data quality and accessibility. [Here](#)
 - b) Collaborations with Educational Institutions: SDAIA has partnered with universities such as King Abdullah University of Science and Technology (KAUST) to establish centers of excellence in data science and artificial intelligence. These collaborations foster research, development, and innovation in AI, aligning with open science principles by promoting knowledge transfer and the development of data-driven solutions. [Here](#)
 - c) Public Engagement Initiatives: Through platforms like the National Data Governance Platform, SDAIA enables public entities and individuals to access and share data securely. This initiative supports the open engagement of societal actors and facilitates open dialogue with other knowledge systems, contributing to the broader open science ecosystem. [Here](#).
 4. TVTC is currently developing policies and strategies to promote open science in alignment with the 2021 UNESCO Recommendation. These efforts aim to foster collaboration with research institutions, universities, and other scientific organizations. Key initiatives under development include:
 - a) Collaborating with Research Institutions: Establishing partnerships with universities and research-performing organizations to align research practices with open science principles.
 - b) Developing an Open Science Policy Framework: Working with the Research, Development, and Innovation Authority (RDIA) to draft guidelines that support open access to research outputs.
 - c) Encouraging Research and Innovation: Creating opportunities for researchers to engage in collaborative and transparent research

projects through initiatives like the Governor's Award for Applied Research.

- d) Promoting Awareness: Organizing workshops and training sessions to raise awareness of open science values and practices across educational and research entities.

- 5. WTIIRA has a strategy that include a goal to building its ecosystem in line with the principles of open science. At a more tactical level this includes:

- a) University Collaboration for Talent and Training.
- b) Government Grants.
- c) Investor Engagement and Financing.
- d) Strategic Partnerships with Saudi Industries and Regulatory Advocacy and Missions.

- 6. Elm Company has a plan to develop and implement open science policies that reflect its dedication to responsible research practices and contributing to the advancement of scientific knowledge.

- 7. JU is committed to creating a strong foundation for open science through these policies, enabling equitable collaboration, promoting innovation, and aligning with global best practices. Accordingly, the University is in the process of developing policies and strategies to promote open science in line with the 2021 UNESCO Recommendation, focusing on fostering a culture of transparency, collaboration, and inclusivity in scientific research. These policies are being designed to address elements of open science, including:

- a) Citizen and participatory science to engage the local community in research activities through citizen science projects.
- b) Co-Production of Knowledge to develop models for collaboration between researchers, such as industry, community organizations, and policymakers.
- c) Recognition of Non-Scientific Collaborations to establish guidelines to recognize contributions from non-scientific collaborators in research projects.

These policies are aligned with the Recommendations:

- a) Encouraging Research-Performing Institutions (17.d): The policy will encourage all university departments and research centers, especially those receiving public funds, to adopt open science practices.
- b) Adopting Statements of Principle (17.e): The University is working on adopting open science principles in coordination with national science organizations, early-career researcher associations, and international bodies like the International Science Council (ISC).

- c) Co-Production of Knowledge (17.g): Establishing frameworks to enable collaboration across disciplines and sectors, ensuring equitable participation and recognition of diverse stakeholders.

These policies and strategies are expected to be finalized and implemented by 2025.

- 8. KAUST was the first university in the region to adopt a mandated Open Access Policy for research publications in 2014. This policy ensures that scholarly articles authored by KAUST researchers are made publicly available, enhancing the visibility and accessibility of our research globally. With over 85% levels of compliance, KAUST is the second among the top universities in the world in terms of compliance level. Over 4M downloads globally through institutional repository.

- a) [KAUST Thesis and Dissertation Archiving Policy, 2010](#)
- b) [Open Access Policy for Scholarly Research Publications](#)
- c) [KAUST Research Data Sharing policy](#)
- d) [Research Data Management Planning](#)

- 9. KAU has implemented strategies and platforms that align with the 2021 UNESCO Recommendation on Open Science, aiming to promote inclusive and participatory access to scientific knowledge.
- 10. The SEU research policy encourages open-access publishing and allows faculty to deposit their research outputs into public repositories. Specific actions also include providing workshops and resources to educate researchers on open science practices.

2.6 In your country, are there specific funding mechanisms for open science at the national level?

Dedicated funding mechanisms for open science have been considered at the national level, including but not limited to:

- 1. RDIA established a unique research grants and funding portfolio that ensures a sustainable and world-class national RDI ecosystem including:
 - a) Saudi Basic Science Program: This program supports fundamental scientific research and encourages the dissemination of research findings through open-access publications, fostering knowledge sharing and collaboration.
 - b) Saudi Applied Research and Technology Program: Aimed at accelerating applied research innovations, this program promotes

- partnerships between academia and industry, emphasizing the open exchange of knowledge and technological advancements.
- c) Saudi Emerging Investigator Program: Focused on nurturing early-career researchers, this program supports initial research projects with an emphasis on open methodologies and data sharing to build a foundation of transparency in research practices.
2. SDAIA is actively working on establishing dedicated funding mechanisms to support open science initiatives. This includes potential grants for open data projects and collaborative research programs focused on data accessibility and AI-driven solutions. These efforts align with the broader objectives of the National Strategy for Data and Artificial Intelligence, which emphasizes fostering innovation through financial support and resource allocation. [Here](#).
 3. TVTC, in coordination with the Research, Development, and Innovation Authority (RDIA), is actively exploring funding mechanisms to support open science at the national level. Current efforts under development include:
 - a) Integrating Open Science into Funding Programs: Proposals are being designed to incorporate open science requirements into publicly funded research grants.
 - b) Collaborating with National Entities: Joint efforts with RDIA and other stakeholders aim to establish dedicated funding streams for open science initiatives.
 - c) Supporting Infrastructure Development: Funding is being considered to enhance open science infrastructure, including data repositories and collaborative platforms.
 - d) Encouraging Research Transparency: Financial incentives are being proposed for researchers who adopt open science practices, such as publishing in open-access journals and sharing datasets.

These mechanisms are part of broader efforts to align funding policies with the values and principles of open science as outlined in the 2021 UNESCO Recommendation.

4. WTIRA/SWA hosts ~2 to 3 times/year event named "Water Research Community" where we try to gather local and international universities to share their works and discuss common interests. SWA funds selected proposals from these events and in certain cases covers the cost of open-access publications. The WTIRA strategy is also fully aligned with national interests and through a mission-driven operating model it will also expect to receive funding from the RDIA grant scheme through SWIC is also establishing a venturing model to take technologies to impact.

5. KACARE has several funding mechanisms through the support provided by the running training program and the Sectorial Portal of Human Capacity Development Building, and the Renewable Energy Atlas Portal by the Renewable Energy Sector. These programs fund studentships for training and development in the energy field. Additionally, they offer grants to support educational and technological institutes and research centers. Also, they support open science innovation. Renewable Energy Incubators and Hackathons provide platforms for innovation and entrepreneurship by fostering collaboration between experts, researchers, and entrepreneurs, supporting open science through knowledge sharing, offering resources, and connecting with international communities, while also contributing to the empowerment of national capacities and the enhancement of local content in renewable and atomic energy fields.
6. IDS has allocated a budget to publish the institute's publications and to reward participants.
7. The ERI Foundation at NEOM funds research activities that lead to publications of the findings in open-access journals and accessible institutional repositories.
8. NeLC has specific funding mechanisms to support open science at the national level. NeLC collaborates with various partners, including public and private entities, to promote the sharing of open educational resources (OER). Through the OERx platform, over 33,000 open educational resources have been shared, and contributed by entities such as ministries, public universities, and private institutions. This collaboration is supported by quality assurance standards designed to empower entities to confidently share high-quality resources.
9. In terms of IP, the National IP Strategy, launched by the Crown Prince in December 2022, is a comprehensive national initiative designed to indirectly support the development of outputs from research and development (R&D) and universities. [Link](#). In addition the National Network of IP Support Centers was established to enable innovators to obtain technical information in the fields of intellectual property and facilitate their access to related services to encourage the investment of their innovative potential and the development, protection, and management of their intellectual property rights. [link](#)
10. KSGAAL is currently exploring opportunities to fund open science research, especially as a non-profit organization.
11. The KSSO relies on institutional and partnership funding rather than dedicated mechanisms for open science. Plans are being explored to collaborate with funding agencies to support open science initiatives.

12.The SEU introduces [Scientific Research Incentives](#) to encourage open-access publication by compensating researchers for open-access publication fees.

13.The QU encourages and supports research to opt for open access to promote open science and knowledge through different means including covering the open access publication fees, etc. The Directorate of Publications and Translation holds the responsibility to provide support for faculty staff for:

- a) Book Publication
- b) Book translation
- c) Hosting several local journals that are publicly available with complete free access.

Also, it is a requirement and part of the promotion process at the QU to show approval of providing social services and knowledge transfer to the national communities.

14.JU, in collaboration with national authorities such as the Council of Universities Affairs and the Research, Development, and Innovation Authority (RDIA), is working to establish funding mechanisms for open science at the national level. These mechanisms aim to foster equitable public-private partnerships, encourage innovative approaches to scientific collaboration, and invest in advanced data science education and stewardship. Details of funding mechanisms under development:

a) Public-Private Partnerships (17.i):

- Objective: To create equitable partnerships with the private sector while ensuring appropriate certification.
- Implementation: Partnerships will align public and private interests to support open science projects and ensure public benefit from publicly funded scientific activities.

b) Innovative Approaches (18.i):

- Objective: To encourage innovation throughout the scientific process and foster international scientific collaboration.
- Implementation: Funding for interdisciplinary projects that integrate machine learning, artificial intelligence, and data visualization, and supporting collaborations between local and international research institutions to enhance knowledge sharing and innovation.

c) Data Science Education and Stewardship (19.c):

- Objective: To invest in advanced education and professionalization in data science, focusing on roles like data governors and stewards to ensure strategic data management and openness.

- Implementation: developing training programs for data stewardship in line with FAIR (Findable, Accessible, Interoperable, Reusable) and CARE (Collective Benefit, Authority to Control, Responsibility, Ethics) principles, and allocating funding to research institutions for hiring skilled data professionals and building infrastructure for secure and open data management.

The funding mechanisms are expected to be finalized and operational by 2025, in cooperation with RDIA. Al-Jouf University is committed to finalizing these funding mechanisms to enable open science practices and support national and global scientific advancement, by adopting the following planned actions:

- a) Establishing pilot projects to test public-private partnerships in open science.
- b) Launching training programs for researchers and data professionals on open science and data management practices.
- c) Securing national and international funding to support innovative scientific collaboration.

2.7 In your country, have open science practices been integrated into existing research funding mechanisms?

Yes, several national organizations have integrated open science practices in their existing funding mechanisms or considered such actions to promote the open science culture and understanding among their communities.

1. RDIA has integrated open science practices into its research funding mechanisms to promote transparency, accessibility, and collaboration within Saudi Arabia's research ecosystem.
 - a) Open Access to Government-Funded Research Infrastructure Policy aims to facilitate external access to government-funded research facilities, optimizing their use and promoting open innovation. The policy mandates that, where practicable, all government-funded research infrastructures in Saudi Arabia should be made available for open access, regardless of the managing host organization. This approach ensures that research facilities are utilized efficiently and are accessible to a broader range of researchers and innovators.
 - b) Research Grants Incorporating Open Science Principles:
 - Saudi Basic Science Program: This program supports fundamental scientific research and encourages the dissemination of research findings through open-access publications, fostering knowledge sharing and collaboration.
 - Saudi Applied Research and Technology Program: Aimed at accelerating applied research innovations, this program promotes

partnerships between academia and industry, emphasizing the open exchange of knowledge and technological advancements.

- Saudi Emerging Investigator Program: Focused on nurturing early-career researchers, this program supports initial research projects with an emphasis on open methodologies and data sharing to build a foundation of transparency in research practices.

These programs are structured to support various stages of research, development, and innovation, aligning with national priorities such as Health and Wellness, Sustainable Environment and Supply of Essential Needs, Energy and Industrial Leadership, and Economies of the Future. [Link](#).

2. TVTC is in the process of integrating open science practices into existing research funding mechanisms in collaboration with national stakeholders. Current efforts include:
 - a) Aligning Funding Policies with Open Science: Discussions are underway with RDIA to incorporate open science requirements, such as open access to publications and datasets, into research grants.
 - b) Promoting Open Access: Proposed guidelines will encourage researchers receiving public funding to publish their findings on open-access platforms.
 - c) Enhancing Collaboration: Funding mechanisms are being adapted to prioritize projects that emphasize transparency, data sharing, and collaboration among researchers and societal stakeholders.
 - d) Building Awareness: Efforts are being made to educate researchers and institutions about the benefits of integrating open science principles into funded projects.

These initiatives aim to establish a framework for embedding open science practices into research funding at the organizational level.

3. The SWA has an STI unit responsible for digitization and hosting published data. The majority of this information is shared through our website, in Arabic and English language.
<https://www.swa.gov.sa/en/about-institute>
<https://www.swa.gov.sa/en/reports>
4. KACARE adopts the Universities Collaboration Program that promotes collaboration work with universities on scientific knowledge sharing, open access publications, innovation research, Intellectual Properties, and open use for research and technical infrastructures and tools.
5. NEOM's research findings are published in open-access journals and institutional repositories and are presented at national and international conferences and workshops.

6. Saudi Standards, Metrology and Quality Organization (SASO) publishes all of its research on its official website. [Link](#).
7. All research funded or conducted by NeLC whether internally or in collaboration with national or international partners—we require that the research findings be published and made accessible to the public. Additionally, for certain research projects, we mandate a dissemination component, where researchers present their findings to relevant stakeholders, fostering transparency and broadening the impact of the research. [Studies and Report](#).
8. KSGAAL is currently engaging in some open science practices, though unintentionally, by providing free access to a wealth of resources, including our journals, data, and materials. We share research data, software, and tools, and we provide infrastructure funding to support research. Additionally, all funded research is required to be produced under an open-source policy. However, it is worth noting that while these practices align with open science principles, their adoption was not initially intentional, as we were unaware of the formal framework at the time. Also, KSGAAL has many open-access libraries and journals:
 - <https://library.ksaa.gov.sa/index/book>
 - <https://search.mandumah.com/>
 - [link](#)
9. MODON – Saudi Authority for Industrial Cities and Technology Zones, aligned with the National Policy published by SDAIA, has provided an open data library on its official website that allows visitors, interested parties, and researchers to obtain general industrial data. [Link](#). In addition, MODON provides a page for requesting the public data [Link](#).
10. The KFNL offers its publications through free open access to all segments of society believing in the importance of knowledge and its role in the development of the individual and society.
11. IDS invites Saudi and non-Saudi researchers, faculty members, experts, and writers inside or outside Saudi Arabia to publish their work in our publications.
12. The KAWLA complex is developing several open source works based on facilitating the practice of open science by providing their own digital repositories to share some of the scientific publications they own, or at least make their data available.

13. NeLC has developed quality assurance standards for licensed entities and online programs, one of which mandates that the institution allows an adequate percentage of its digital content to be made available as open resources. To support this standard, platforms like OERx are encouraged to enrich digital content and contribute to the development of community members. This aligns with the values of open science, such as equity, inclusiveness, and collective benefit, and the guiding principles of transparency, collaboration, and sustainability.

14. KAPSARC has general guidelines and 'best practices' in its operational guidelines that are more relevant to open science to scientific knowledge, including the following:

Section 5 Publishing/General guidelines

"h) Open Access

In general, all KAPSARC research publications will be made publicly accessible and free to reuse under the sole condition that the source is properly cited.

- Research articles authored or co-authored by KAPSARC personnel that are published in external peer-reviewed literature.

Authors of articles or other publications such as books, being published by external parties, should connect with the KAPSARC K&A Library to organize Open Access arrangements, publishing permissions on the KAPSARC website and any other required licenses. In the case of models, data, and tools, the SP team shall publish following the Intellectual Property guidelines of KAPSARC."

Section 5.2 Publishing a research paper externally.

"c) The K&A Library is responsible for arranging Open Access for papers published externally."

15. JU has integrated open science practices into existing research funding mechanisms, as the university provides support for researchers to publish in Open Access journals by reimbursing publication fees to encourage the adoption of these practices, which contributes to making scientific knowledge widely accessible.

16. KAUST has integrated the open access policy implementation into the faculty's annual evaluation process. The evaluation section, which is related to publications and research outputs, is tied to repository submissions. For a faculty member's publication to be included in the end-of-year report, the publication record must exist in the institutional repository.

2.8 Have efforts been made or are foreseen to be undertaken by the end of 2025 in your country to foster equitable public-private partnerships on open science in line with the 2021 Recommendation?

Yes, there are several considerable efforts that have been made to foster equitable public-private partnerships on open science. Some of these efforts are matured more than others depending on the relevance of open science to the mandated responsibilities at the organizational level, as follows.

1. RDIA is actively fostering equitable public-private partnerships to promote Open Science in alignment with the 2021 UNESCO Recommendation on Open Science. These efforts are designed to enhance collaboration, transparency, and accessibility in scientific research across Saudi Arabia.

Key Initiatives:

- a) Open Access to Government-Funded Research Infrastructure Policy:
Objective: To facilitate external access to government-funded research facilities, optimizing their use and promoting open innovation.
Implementation: The policy mandates that, where practicable, all government-funded research infrastructures in Saudi Arabia should be made available for open access, regardless of the managing host organization. This approach ensures that research facilities are utilized efficiently and are accessible to a broader range of researchers and innovators.
- b) Assessment of Academic-Corporate Partnerships:
Objective: To evaluate and enhance collaboration between academia and industry, identifying best practices and areas for improvement.
Implementation: RDIA has conducted assessments to understand the current landscape of academic-corporate partnerships in Saudi Arabia. These assessments aim to reduce barriers to innovation and cultivate a supportive regulatory environment, thereby promoting equitable partnerships.
- c) National Semiconductor Initiative:
Objective: To strengthen Saudi Arabia's semiconductor sector through advanced training programs and localization of digital chip design technologies.
Implementation: RDIA, in collaboration with the National Academy for Industry and other governmental bodies, has launched training programs aimed at equipping national talent with specialized knowledge in semiconductor design. This initiative fosters public-private partnerships by preparing participants to contribute to electronic chip design projects within various companies, thereby advancing the semiconductor industry in Saudi Arabia.

Through these initiatives, RDIA demonstrates its commitment to fostering equitable public-private partnerships in Open Science, thereby enhancing the collaborative and innovative landscape of scientific research in Saudi Arabia.

2. MOE Madares platform contains information about private schools in Saudi Arabia, enabling parents to choose a school based on certain criteria such as geographic location, tuition fees, financing solutions, and other features. <https://tetco.sa/ar/madares>
3. SDAIA is actively fostering public-private partnerships to support open science initiatives. Key examples include:
 - a) Collaboration with Industry Leaders: Partnerships with companies such as Microsoft and Google to provide access to advanced AI tools and cloud infrastructure, enhancing data sharing and open-access research projects.
 - b) Joint Initiatives with Tech Startups: Engaging with local startups to co-develop AI solutions that are shared publicly, fostering innovation while adhering to open science principles.
 - c) Support for Open Data Projects: Launching joint projects with private sector entities to enhance data accessibility through the National Data Bank, promoting transparency and open collaboration.

These efforts align with the 2021 UNESCO Recommendation by encouraging equitable access to resources and fostering a collaborative environment for open science.

4. WTIRA/SWA continues to expand its Water Research Community events and funding through this event while offering more sources for open-access publications. In addition, SWIC will continue to evolve its venturing model for operationalizing technology for widespread use, representing mature applications.
5. KACARE has some projects in the technology localization initiative that involve public-private partnerships to develop technologies in renewable energy.
6. Efforts have been made at JU, and additional initiatives are planned to be undertaken by Al-Jouf University by the end of 2025 to foster equitable public-private partnerships (PPPs) on open science, aligning with the 2021 UNESCO Recommendation. [Link](#)

Objectives:

- Ensure the private sector contributes to and benefits from open science initiatives in a manner aligned with the global public interest.

- Prevent market dominance by commercial entities while promoting fair and ethical practices.
- a) Current Efforts:
 - Collaborations with Industry: Al-Jouf University has initiated partnerships with private sector entities to support research projects that emphasize open science principles.
 - Joint Research Projects: Public-private collaborative research projects are being designed to ensure shared benefits and equitable participation.
- b) Planned Initiatives:
 - Development of Certification and Regulation Frameworks:
 - Workshops and Dialogue Forums:
 - Hosting events to engage private sector stakeholders in discussions on open science principles, benefits, and shared responsibilities.
- c) Expected Outcomes by 2025:
 - Strengthened collaboration between public and private sectors in research.
 - Development of a transparent framework ensuring fair contributions and benefits for all stakeholders.
 - Enhanced private sector engagement in national and global open science goals.

These efforts align with JU's mission to foster transparency, innovation, and equitable partnerships in scientific research.

7. National Gard has some efforts to foster equitable public-private partnerships, including partnerships with pharmaceutical and biotechnology companies. [Link](#).
8. Nelc has made efforts to foster equitable public-private partnerships in open science through the FutureX platform, in alignment with UNESCO's 2021 recommendations. Key initiatives include:
 - a) Partnership with Coursera: Through the FutureX platform, Nelc established a partnership with Coursera to provide high-quality courses and educational programs for Saudi citizens, students, and government employees, enhancing digital learning opportunities and open access to educational resources.
 - b) Partnership with edX: In October 2021, Nelc collaborated with edX via the FutureX platform to offer Saudi citizens and government employees broad access to world-class courses and educational programs, strengthening open access to educational resources and knowledge sharing.
 - c) Partnership with ICESCO: In June 2021, Nelc signed an agreement with the Islamic World Educational, Scientific, and Cultural Organization

(ICESCO) to promote eLearning among member states. This partnership aims to exchange expertise and support open science practices across Arab countries, fostering international collaboration and the dissemination of open scientific knowledge.

9. Efforts are being made and are planned at MODON to foster equitable public-private partnerships, particularly through data sharing and open data initiatives. These efforts include encouraging collaboration between public and private sectors to enhance transparency, innovation, and accessibility of scientific data while ensuring equitable benefits for all stakeholders.
10. KSGAAL sets a plan with our partners (STC and PSU) to open labs to provide opportunities for the researchers.
11. HADITH aims to develop specialized scholarly content and will subsequently work on strengthening partnerships between the public and private sectors.
12. KSSO collaborates with private sector companies to develop exhibitions and educational programs that reflect open science principles, fostering broader societal engagement.
13. The Islamic University has made efforts or are foreseen to be undertaken by the end of 2025 to foster equitable public-private partnerships on open science in line with the 2021 Recommendation. These efforts include:
 - a) Policy Development: Working on developing policies that support partnerships between public and private sectors in open science, enhancing collaboration between academic institutions and private companies.
 - b) Workshops and Training: Organizing workshops aimed at policymakers and professionals in the field to raise awareness of the importance of public-private partnerships in promoting open access to knowledge and data.
 - c) Global Partnerships: Engaging in global partnerships such as the Global Open Science Partnership, which supports collaboration between publishers and funding agencies, promoting the exchange of knowledge and best practices.
 - d) Knowledge Exchange: Creating platforms for knowledge exchange among various stakeholders, including academic institutions, the private sector, and civil society, to enhance mutual understanding of the benefits of open science.

- e) Progress Evaluation: Establishing mechanisms to evaluate progress in implementing public-private partnerships in open science, helping to identify gaps and challenges that need to be addressed.

2.9 Is there a national monitoring framework for open science in your country? Or are there specific indicators of open science included in the national monitoring and evaluation framework for STI policy?

Saudi Arabia has set a National Data Governance Policy (2020) developed by the Saudi Data and AI Authority (SDAIA), aiming to govern national data, and set policies on data classification, sharing, data collection and processing, and the freedom of information. SDAIA, also, established a [National Data Governance Platform](#), which is a national electronic Platform that targets data management and governance in addition to the protection of personal data by providing a set of electronic services, such as: Assessing compliance, receiving and addressing complaints and inquiries related to violations of PDPL to protect data as national assets and safeguard individuals' rights from abuses and illegal violations based on PDPL, issued by Royal Decree No. (M/19) dated 09/02/1443 AH, and amended by Royal Decree No. (M/148) dated 05/09/1444 AH. Moreover, via the Platform, beneficiaries from the public, private, and non-profit sectors and individuals are supported with tools, guides, and advisory services that enable them to comply. The main objectives of the National Data Governance Platform are:

- a) Maintain the privacy of individuals and enable them to exercise their rights when processing their personal data.
- b) Enhance national data sovereignty and maintain the privacy of individuals when processing their personal data.
- c) Raise awareness of personal data protection and reduce risks of data breaches and violations.
- d) Enable individuals to file complaints against PDPL violators.
- e) Enhance confidence in trading and commercial transactions in the Saudi market through a national data governance that protects clients' rights.

In addition, SDAIA and The National Data Management Office through the National Data Index (NDI) measure the compliance of Open Data publishing. [Link](#).

Other national monitoring frameworks are as follows:

1. The Research, Development, and Innovation Authority (RDIA) has adopted a national monitoring framework for Open Science, aligning with the 2021 UNESCO Recommendation on Open Science. This framework includes specific indicators within the national monitoring and evaluation system for Science, Technology, and Innovation (STI) policy.

National Monitoring Framework for Open Science:

Implementation: The RDIA has integrated Open Science indicators into its STI monitoring framework to assess and promote transparency, accessibility, and collaboration in scientific research.

Key Indicators:

- a) Open Access Publications: Tracking the proportion of publicly funded research outputs available through open access channels.
 - b) Data Sharing Practices: Monitoring the extent to which research data is shared and reused across institutions.
 - c) Collaborative Research Initiatives: Evaluating the number and impact of research projects involving multiple stakeholders, including public and private entities.
2. The Ministry of Education is working to make the following initiatives/projects as open access platforms/sources:
 - a) Awareness and Cultural Adaptation Program for Parents.
 - b) Development of Distance Learning Frameworks.
 - c) Attracting Global Higher and Vocational Education Institutions to the Kingdom.
 - d) Higher Education for Everyone.
3. TVTC is currently collaborating with national stakeholders, including RDIA, to establish a national monitoring framework for open science. Efforts under development include:
 - a) Defining Key Performance Indicators (KPIs): Proposed indicators will measure open science practices, such as open-access publications, data sharing, and collaborative research.
 - b) Aligning with STI Policies: The monitoring framework will be integrated with the broader Science, Technology, and Innovation (STI) policy evaluation mechanisms to ensure consistency and alignment with national goals.
 - c) Collaborative Frameworks: Discussions are ongoing to develop a unified system for monitoring open science initiatives across research institutions and organizations.

- d) Capacity Building: Training programs and workshops are being organized to ensure proper understanding and implementation of monitoring mechanisms among stakeholders.
4. NeLC applies a national monitoring framework for open science through the [National Digital Learning Indicator](#). This framework includes specific indicators related to open science, such as the number of open resources provided by entities.
 5. WTIRA measures its progress against its strategic goals, one of which is building our ecosystem with strategies and tactics. The WTIRA understands that Saudi Arabia is making strides towards embracing open science and the establishment of a dedicated national monitoring framework for open science is still in progress.
 6. The Saudi Authority for Intellectual Property (SAIP) supports open science by promoting policies that encourage the registration and management of intellectual assets, fostering innovation and collaboration while ensuring compliance with intellectual property regulations. [link](#)
 7. Al-Jouf University is currently in the process of developing a framework to support and monitor open science. Existing efforts include promoting open access by reimbursing publication fees for researchers, which serves as an initial step toward fostering open science and developing more comprehensive policies.
 - a) Description of Current Support:
 - The university supports researchers by reimbursing publication costs in open-access journals, reducing financial barriers, and encouraging open-access publishing.
 - b) Goals of the Framework Under Development:
 - Establishing clear policies to support open science, including open access to research and data, promoting collaboration among researchers, and enhancing transparency.
 - Developing monitoring indicators to assess open science efforts, such as the number of open access publications, allocated funding, and levels of public engagement in research dissemination.
 - c) Implementation and Oversight Mechanisms:
 - Creating specialized committees or units to oversee the implementation of open science policies.
 - Involving multiple stakeholders, such as researchers, academic departments, and public and private sector partners, to ensure a holistic framework.
 - d) Future Plans:

- Developing infrastructure to support open science, such as open databases, digital repositories for research outputs, and educational platforms.
 - Ensuring the sustainability of financial support for publication reimbursement and maintaining transparency in the process.
- e) General Oversight and Inclusivity:
- The framework will be designed under public oversight, ensuring broad participation from the academic community and adherence to principles of transparency and fairness.

3. INVESTING IN OPEN SCIENCE INFRASTRUCTURES AND SERVICES

3.1. What is the latest official data for your country's percentage of national gross domestic product (GDP) dedicated to research and development expenditure? Please also indicate the year of publication of this data, and to which year(s) it refers.

Research and Development (R&D) expenditure is a vital metric reflecting a country's commitment to scientific progress, technological innovation, and economic diversification. In Saudi Arabia, R&D has become an essential pillar of the Kingdom's Vision 2030, aiming to diversify the economy, enhance technological capabilities, and reduce reliance on oil revenues. In 2022, Saudi Arabia allocated 0.46% of its GDP to R&D [sa.pdf](#). As the Kingdom aims to increase its R&D expenditure to 2.5% of GDP by 2040, the future looks promising for Saudi Arabia's continued growth as a regional leader in scientific research, technological innovation, and economic diversification. The rising investment in human capital and the growing role of the private sector will be key drivers of this transformation.

3.2. What is the latest official data for the percentage of your country's population that has access to reliable internet and bandwidth? Please also indicate the year of publication of this data, and to which year(s) it refers.

Saudi Arabia's internet accessibility and bandwidth availability have made impressive strides, with the country now boasting an internet penetration rate of 99%, in 2023 ([Link](#)). This is a clear reflection of the nation's commitment to digital transformation and technological advancement under the Vision 2030 framework. The high-speed internet infrastructure available across the Kingdom supports not only the daily lives of citizens but also the growth of key sectors such as education, research, business, and innovation. As the country continues to roll out 5G and expand fiber-optic networks, Saudi Arabia is well-positioned to be a regional leader in digital connectivity and bandwidth capacity.

3.3. Are there national research and education networks (NRENs) active in the country?

Yes, Saudi Arabia develops National Research and Education Networks (NRENs) to provide high-speed connectivity and collaboration platforms for academic and research institutions. These networks play a critical role in supporting research initiatives, fostering innovation, and promoting the exchange of knowledge across national and international boundaries. In Saudi Arabia, efforts to establish and improve such networks align with the nation's Vision 2030 objectives to transform the country into a knowledge-driven economy.

The Saudi Research and Innovation Network (Maeen) is the most prominent network in Saudi Arabia. [link](#). Maeen Network is a national research network parallel to the current Internet, established by the King Abdulaziz City for Science & Technology (KACST) in December 2017. It was developed with national expertise in collaboration with national research and education institutes, in line with Saudi Vision 2030. Maeen Network is characterized by its speed and efficiency in transferring data between its members, allowing the benefit of various applications and services to meet the needs of government and research agencies in the scientific research community across various geographical regions. It serves as an enabler for various goals related to the Education and Research domains. It provides unique services and creates a conducive environment for academics to conduct research and test innovative technologies, thereby enhancing knowledge sharing and education development within the Kingdom. The network connects academic entities within the country and links to peer networks around the globe, offering unique global services. Also, Maeen Network builds distinguished partnerships with similar scientific research and education networks worldwide. It is interconnected with the global Internet2 network, established by higher education institutes in the United States of America, characterized by high speeds that reach several times the data transfer speed of the regular Internet. This connection provides Maeen Network subscribers access to more than 90,000 academic institutions, 42 educational networks, and 65 research networks, representing over 100 countries around the world. Additionally, the Maeen Network is interconnected with the European Research Network (GÉANT) through dedicated 10 Gbps links extending to the NetherLight data center in the Netherlands. GÉANT is one of the world's largest and most influential research networks, interconnecting all research networks in Europe. This facilitates faster and easier data exchange between Maeen and other European research networks.

Other active national research and education networks (NRENs) in the country are:

- a) Unified Arab Index (affiliated with King Abdulaziz Public Library)
- b) King Faisal Center for Research and Islamic Studies.
- c) General Presidency for Scholarly Research and Ifta.
- d) [Saudi Digital Library \(SDL\)](#).
- e) [National eLearning Center \(NELC\)](#).
- f) [Open Educational Resources \(OERX\)](#)
- g) The platform of the Cultural Skills Competition [link](#).
- h) The Future Entrepreneur Platform in cooperation with Monsha'at for entrepreneurship programs, includes open educational videos, as well as open and announced data for the results of the interactions of all education departments in the competition. [link](#)

3.4. Are there national or institutional open science infrastructures in your country?

Yes, Saudi Arabia is making significant progress in the development of open science infrastructures at both national and institutional levels. Several institutions and government entities are taking proactive steps toward fostering an open science culture.

1. KACST serves as the foremost open science infrastructure at the national level, functioning as the reference national laboratory center across scientific, technological, and research domains. It is a key reference point for the transfer, localization, and development of exclusive technologies, contributing to the enhancement of local content. KACST effectively bridges the gap between scientific research and industrial technology application, while also striving to achieve global competence and facilitate technology transfer. Additionally, it hosts a science and technology park aimed at expediting the commercialization of research and development activities. The organization has established its 2030 Strategic Direction, which integrates pertinent initiatives aligned with open science principles.
2. RDIA supports Saudi Arabia's open science infrastructure by promoting collaboration through platforms like Maeen, [GESR](#), and [Saudi Minds](#). These initiatives enhance connectivity, regional research collaboration, and access to vital resources, aligning with Vision 2030's goals of fostering a knowledge-driven economy and advancing innovation.
3. Nelc has developed several open science infrastructures in alignment with UNESCO's 2021 Recommendation on Open Science. These include:
 - a) Federated Information Technology Infrastructure for Open Science:
 - NeLC provides cloud computing and data storage solutions supporting open educational resources (OER) and high-capacity data exchange.
 - FutureX is an example of the infrastructure NeLC uses to support data storage, computing, and resource access.
 - Compliance: These infrastructures are designed for inclusivity, accessibility, and interoperability, promoting FAIR principles to ensure the resources are accessible, interoperable, and reusable.
 - b) Platforms for Exchanges and Co-Creation of Knowledge Between Scientists and Society:
 - NeLC's [OERx platform](#) enables collaborative learning and access to educational content, fostering an open science culture where resources are shared between institutions, researchers, and society.

- Compliance: This platform is non-commercial, community-governed, and adheres to FAIR and CARE principles, making educational resources accessible, reusable, and ethically managed.
4. SDAIA has established several open science infrastructures as part of its strategic initiatives:
 - a) National Data Bank (NDB): The NDB consolidates data from multiple government entities and provides a central repository for open data access. It supports data sharing and transparency across various research domains.
 - b) Data Marketplace: A platform integrated with the NDB that facilitates secure data sharing between government entities, private sectors, and researchers. It aligns with open science principles by promoting accessibility and collaboration.
 - c) Collaborative Data Labs: These labs provide physical and virtual spaces for collaboration on data-driven projects, supporting the co-creation of knowledge and public engagement.
 - d) High-Performance Computing Infrastructure: SDAIA has invested in high-performance computing resources to enable large-scale data analysis and support open-access research in AI and data science
 6. KSGAAL contributes to Saudi Arabia's open science infrastructure by engaging with platforms like [ARAI](#) and [AlMustashar](#), fostering collaboration across academic, government, and private sectors. These efforts support research in Arabic language and linguistics, promoting knowledge exchange and innovation in alignment with Saudi Arabia's Vision 2030 goals for a knowledge-driven economy.
 7. KACARE develops Open Science Infrastructures including:
 - a) Renewable Energy Resources (Atlas) Portal and Physical IoT Devices ([Link](#))
 Description: Provides detailed geospatial and temporal data on renewable energy resources across Saudi Arabia. Integrated with IoT devices for ground-based monitoring. The monthly resolution data is open to the public; national and international. The Atlas complies with the 2021 Recommendation:
 - Inclusivity & Accessibility: Designed for public, private, and institutional access, including researchers, investors, and policymakers.
 - Interoperability: Data provided in standardized formats compatible with global energy and meteorological systems.
 - FAIR Principles: Data is findable via metadata standards, accessible through an online portal, interoperable with scientific software, and reusable.

- CARE Principles: Ensures ethical use of energy data to promote sustainable development while safeguarding local energy interests.

Commercial Status: The monthly resolution data is non-commercial and open to the public; national and international. The higher resolutions are commercial.

b) Technology Local Content Portal ([link](#))

Description: Links technology providers with local suppliers and facilitates application to the technology localization initiative. The Portal complies with the 2021 Recommendation:

- Inclusivity: Provides a platform for local Content technologies to allow developments.
- Accessibility: Open to researchers and industries globally, in partnership with Saudi entities.
- Interoperability: Aligns with regional and international innovation standards.
- FAIR/CARE: Emphasizes equitable benefits for local communities and ethical data use.

Commercial Status: Non-commercial.

c) Studies and Research Devices and Tools

Description: Offers access to specialized tools for research and education in energy and data science. Includes measurement devices, simulation software, and analytical resources. These devices and Tools comply with the 2021 Recommendation:

- Inclusivity: Supports a wide range of researchers, including students and academics.
- Interoperability: Ensures compatibility with Energy data analytical tools globally.
- FAIR/CARE: Adheres to guidelines ensuring data accessibility and ethical handling of intellectual property.
- Commercial Status: Non-commercial.

Link: KACARE – Studies and research site.

d) The Knowledge Resources Portal

Description: A comprehensive repository for research outputs, educational materials, and policy documents related to the energy and sustainability domain. The Portal complies with the 2021 Recommendation:

- Inclusivity: Targets a broad audience, including global researchers, students, subject matter experts, and the general public.
- Accessibility: Freely available to all users.
- Interoperability: Integrated with global knowledge platforms using open standards.

- FAIR/CARE: Promotes reusability of resources while ensuring ethical use and community governance.
 - Commercial Status: Non-commercial.
 - Link: Under Development
8. IDS significantly contributes to advancing research in diplomatic and international studies in Saudi Arabia by enhancing access to resources and fostering collaboration. Through its engagement with platforms like International Studies and The Diplomat List, the Institute supports a knowledge-driven economy, aligns with Vision 2030, and promotes global partnerships and open science infrastructures.
 9. NEOM is enhancing its research capabilities by integrating Elsevier's PURE Research Management System, strengthening its open science infrastructure, and promoting global collaboration, data sharing, and efficient research management. This aligns with Saudi Arabia's Vision 2030 goals by advancing scientific innovation and supporting sustainable development initiatives.

3.5. Does your country host or fund any federated regional or international information technology infrastructure for open science?

Yes, Open Science initiatives aim to promote the free flow of scientific knowledge by making research data, tools, and educational content more accessible, reusable, and transparent. This requires the development of robust IT infrastructures that can support the shared, collaborative, and open exchange of knowledge across geographical and institutional boundaries. In this context, federated regional or international IT infrastructures for open science play a crucial role in enabling institutions, researchers, and educators to access and contribute to global open science efforts. Saudi Arabia has a remarkable federated regional information technology infrastructure for open science, among the most prominent ones are:

1. [Gulf Information Database \(GESR\)](#)

King Abdulaziz City for Science and Technology (KACST) launched the Gulf Information Database (GESR) in 2023. The launch occurred within the twenty-fifth meeting of the Committee of Presidents and Directors of Universities and Higher Education Institutions of the Gulf Cooperation Council (GCC) and the GCC Secretariat. GESR is hosted, operated, and developed by the Saudi Research & Innovation Network (Maeen Network). GESR aims to establish a unified Gulf electronic information system to inventory scientific production, document research resources in Gulf academic institutions, and make data accessible to researchers. GESR is considered one of the active tools in scientific research in the Arabian Gulf region, and one of the tributaries of information exchange that

provides reliable and value-added services to beneficiaries such as experts, entrepreneurs, individuals, government and research agencies, emerging technical institutions, and the industrial and private sectors. GESR enriches research, development, and innovation institutions, research and study centers, educational institutions, experts and specialists, studies, research, and scientific production in GCC. GESR consists of an interconnected set of digital databases, including, but not limited to, the following:

- a) A specialized database of research, development, and innovation institutions in the GCC countries. This includes universities, research and studies centers, educational institutions, and the private sector.
- b) A specialized database of experts and specialists in the GCC countries, including faculty members, specialists, assistants, researchers, and scholars in universities, research and study centers, and various educational institutions.
- c) A specialized database in the field of studies, research, and scientific production is available in all universities, research centers, study centers, and various educational institutions in the Gulf countries.

GESR is governed by a supervisory committee representing the main research and development bodies in each Gulf country, headed by King Abdulaziz City for Science and Technology (KACST). The Supervisory Committee shall submit its periodic reports to a higher committee chaired by the Secretary General of the Gulf Cooperation Council (GCC).

2. National eLearning Center (NeLC) hosts and funds an integrated information technology infrastructure to support open science at the regional level. This infrastructure includes:

a) Institutional Micro-Sites for Developing Open Educational Content.

- Language: Arabic and English
- Target Groups and Beneficiaries: Educational institutions seeking to develop and share open educational content
- Governance/Community Agreements: Managed by institutions under guidelines provided by NeLC.
- Scope: Research Areas: Varies based on institutional focus Elements of Open Science: Open access and community engagement in content creation
- Stage of Research: Primarily for the dissemination and application of educational content.
- Accessibility: Available locally and regionally, with potential international access based on institutional policies.

b) Digital Repositories:

- Access: Access to digital repositories is provided through the micro-sites, allowing institutions to build their programs and develop courses flexibly.

- Governance: The repositories adhere to ****FAIR**** principles (Findable, Accessible, Interoperable, and Reusable) to ensure resources are easily accessible and ****CARE**** principles (Collective Benefit, Authority to Control, Responsibility, and Ethics) for ethical management and transparency.
 - Non-Commercial: These infrastructures are provided on a non-commercial basis, supporting the dissemination of open educational content.
3. Saudi Water Authority: GPID is to recognize and reward startup scientists, researchers, and individuals making significant contributions to the field of water technologies, as well as to inspire others to join us in our mission to develop sustainable water solutions. The GPID wants to advance the technologies and practices of water technologies to address global water scarcity and improve the lives of people around the world. The prize will support the winner by providing the necessary funding to advance these technologies and the resources needed for their success. In 2024, GPID has received 540 submissions from 80 universities and research institutes across 56 different countries. These submissions span Technology Readiness Levels (TRL) 1 to 3 and 4 to 7. The winner of this year is 14 across both categories [Global Prize for Innovation in Desalination 2024](#).

3.6. Is your country involved in any North-South, North-South-South, and South-South collaborations to optimize infrastructure use and joint strategies for shared, multinational, regional, and national open science platforms?

Yes, Saudi Arabia is increasingly engaging in North-South, North-South-South, and South-South collaborations to enhance the development and use of open science infrastructures. Notable contributions come from organizations such as the National eLearning Center, Saudi Data and AI Authority, KACST, and NEOM, each actively participating in multinational initiatives aimed at optimizing infrastructure and supporting shared open science platforms. These collaborations not only strengthen Saudi Arabia's scientific and technological capacity but also promote knowledge exchange and innovation on a global scale, particularly through partnerships with organizations such as UNESCO, CERN, and the Global Research Council. The Kingdom's continued focus on open science aligns with its Vision 2030 objectives, positioning Saudi Arabia as a key player in the global research landscape.

1. National eLearning Center NeLC actively participates in international partnerships to enhance infrastructure use and develop joint strategies for open science platforms. Notable partnerships include:
 - a) Collaboration with Coursera:
 - Infrastructure Name: FutureX Platform

- Language: Arabic and English
- Target Groups and Beneficiaries: Saudi citizens, students, and government employees
- Governance/Community Agreements: Managed in partnership between NeLC and Coursera
- Scope: Research Areas: Digital education and online learning.
- Elements of Open Science: Open access to educational resources
- Stage of Research: Implementation and dissemination of online courses

b) Collaboration with edX:

- Infrastructure Name: FutureX Platform
- Language: Arabic and English
- Target Groups and Beneficiaries: Saudi citizens, students, and government employees
- Governance/Community Agreements: Managed in partnership between NeLC and edX
- Scope: Research Areas: Digital education and online learning.
- Elements of Open Science: Open access to educational resources.
- Stage of Research: Implementation and dissemination of online courses

c) Partnership with ICESCO (Islamic World Educational, Scientific, and Cultural Organization):

- Infrastructure Name: Joint eLearning Initiatives
- Language: Arabic and English
- Target Groups and Beneficiaries: Member states of ICESCO
- Governance/Community Agreements: Collaborative framework between NeLC and ICESCO
- Scope: Research Areas: eLearning and digital education
- Elements of Open Science: Knowledge sharing and capacity building
- Stage of Research: Development and implementation of eLearning programs.

2. Saudi Data and AI Authority (SDAIA) actively engages in international collaborations to enhance open science platforms and AI infrastructure:

- a) Partnership with UNESCO and IRCAI: SDAIA has collaborated with UNESCO's International Research Centre on Artificial Intelligence (IRCAI) to promote open access AI research and knowledge sharing, supporting North-South and South-South partnerships.
- b) Cooperation with Regional AI Initiatives: Through collaborations with GCC countries and the Arab League AI working group, SDAIA participates in joint projects to develop shared data repositories and AI knowledge exchange platforms, enhancing regional research capabilities.
- c) Global AI Summit Initiatives: As part of the Global AI Summit, SDAIA has facilitated multinational dialogues and partnerships focusing on open

science principles and infrastructure optimization. More information can be found at the summit.

3. KACST plays a key role in advancing open science by participating in regional and international collaborations, such as its involvement with the [Gulf Information Database \(GESR\)](#). This engagement strengthens North-South and South-South collaborations, promotes data exchange, and enhances Saudi Arabia's position in global scientific research, aligning with Vision 2030 goals for innovation and knowledge-sharing.
4. NEOM Initiated partnership exploration with CERN as part of a wider network of KSA organizations that promote higher learning (RDIA, KACST, Effat University, KAUST) to affiliate with CERN as an associate member and to access, for example, CERN's Open Quantum Initiative (an Open Science initiative to create an international network focusing on quantum research). ERI Foundation and NEOM U are engaged in talks with CERN to access CERN's Open Quantum Initiative. That targets users from the South and focuses on global challenges, or challenges with local specificities, such as drought and water usage. Users are partnered with CERN's quantum algorithms group and co-develop algorithms that are tested on quantum computing infrastructure provided by private industry partners such as IBM, Microsoft, and Amazon Web Services. ERI Foundation and NEOM U are additionally exploring a partnership with UNESCO's International Centre for Theoretical Physics (ICTP). ICTP is a Category-1 UNESCO body promoting Open Science, education, and capacity building through global networks of partner research and innovation institutions. ERI Foundation-funded research in collaboration with KAUST has involved collaboration with the Egyptian Environmental Affairs Agency. This work has been published in an open-access journal. Ocean science research, including in-person field work, is on-going in East Timor via the ERI Foundation's Ocean Science and Solutions Applied Research Institute. The results will be published in due course.

4. INVESTING IN HUMAN RESOURCES, TRAINING, EDUCATION, DIGITAL LITERACY, AND CAPACITY BUILDING FOR OPEN SCIENCE.

4.1 Has systematic capacity building on open science taken place in higher education and/or research-performing institutions in your country?

Yes, Saudi Arabia has undertaken a series of systematic capacity-building initiatives focused on open science within higher education and research-performing institutions, that manifested in various forms. These endeavors can be categorized as platforms or academies as follows:

1. The Open Educational Resources Platform (OERx) serves as a national digital hub designed to facilitate the creation, sharing, and utilization of open educational resources. It offers a comprehensive repository of high-quality educational materials, ensuring accessibility for both educators and learners.
2. [FutureX](#) Platform is a national platform that offers access to a wide range of online courses and programs in collaboration with global partners like Coursera and edX. It aims to enhance digital learning and provide flexible learning opportunities.
3. Professional Certification is a professional certification program to develop competencies in e-learning and open educational practices. These programs are designed to equip educators and administrators with the necessary skills to implement and manage open science initiatives effectively. All these efforts are part of NelC initiatives that are created to help in capacity building on open science in higher education.
4. Methodological capacity building in the field of open sciences, represented by implementing training programs and workshops according to the field of specialization of KAWLA, for more information, please visit the following links: <https://www.kawla.gov.sa/ar/events/1483>
<https://www.kawla.gov.sa/ar/events/1483>
https://x.com/kawlaa_sa/status/1853411914690298365
https://x.com/kawlaa_sa/status/1849524718598037695
https://x.com/kawlaa_sa/status/1846888420943057141
https://x.com/kawlaa_sa/status/1844446964307394582
https://x.com/kawlaa_sa/status/1717976017770311929
5. Supporting education in computational linguistics with a strong collaboration with various universities across Saudi Arabia is an important role where King Salman Global Academy for Arabic Language, (King Salman Global Academy for Arabic - KSGAAL) is playing this role.

Additionally, KSGAAL works with many programmers worldwide to train Arabic teachers, enabling them to become qualified to teach Arabic as a second language. Furthermore, we provide opportunities for students from around the globe to come to Saudi Arabia and study Arabic free of charge.

6. Academy 32 (academy32.sa) was established as a hub for cultivating future talents and investing in human potential within the research, development, and innovation sectors. The academy identifies the national needs of leaders, researchers, and qualified specialists. Proposing recommendations and tailored solutions to meet these needs. It offers training, development, and enrichment programs, as well as knowledge activities, following the best practices to build globally competitive citizens.

4.2 Have capacity-building programs for policymakers on how to integrate core values and principles of open science in STI policies and strategies taken place in your country?

Yes, there are several capacity-building programs for policymakers addressing the broad spectrum of open science initiatives in Saudi Arabia. These programs encompass a variety of topics and approaches, including but not limited to the following:

1. The Institute of Public Administration (IPA) plays a crucial role in fostering leadership and governance excellence by providing professional training, research, and consultancy services. It serves as a center for developing administrative skills, policy-making expertise, and ethical leadership among public sector officials, private sector leaders, and aspiring professionals. Through specialized programs, workshops, and capacity-building initiatives, the IPA enhances decision-making abilities, promotes good governance, and encourages innovation in public administration (ref. <https://www.ipa.edu.sa/en>).
2. NEOM (<https://www.neom.com/en-us>) The leadership teams recruited to ERI are highly experienced academics with backgrounds in public institutions of higher learning that practice Open Science so have the instincts that align with the principles of Open Science and the experience of the regulatory frameworks, policies, and strategies to promote them. Detailed discussions on establishing the associated systematic capacity building are taking place. A green paper on building the R&I ecosystem at NEOM addresses establishing the associated R&I culture.

4.3 Has a framework of open science competencies been incorporated into higher education research skills curricula in your country, or is it foreseen by the end of 2025?

Several initiatives aimed at integrating open science competencies into higher education research skills curricula have been implemented in Saudi Arabia. Notable among these initiatives are the following:

1. [OERx](#) Platform actively promotes transparency and collaboration in scientific research integrating open science competencies into higher education research skills.
2. Courses and training on Open Access and related topics are offered by the [Library](#) and KAUST administration regularly (<https://www.kaust.edu.sa>).
3. Relevant initiatives are under development by NEOM (<https://www.neom.com/en-us>). The curricula will include R&I requirements and the culture expected to establish globally ranked R&I activities. They will also require students to participate in research with active research professionals where they will gain first-hand training in Open Science practices.

4.4 Are open educational resources as defined in the 2019 Recommendation on Open Educational Resources used as an instrument for open science capacity building in your country in line with the 2019 Recommendation mentioned above?

Yes, relatively there are some open resources used as an instrument for open science capacity building in your country in line with the 2019 Recommendation, such as:

1. NeLC utilizes open educational resources (OER), as defined in the 2019 Recommendation on OER, as a key instrument for capacity building in open science. Through platforms like "OERx", NeLC provides access to a variety of high-quality, open-access educational resources to support skill development and knowledge sharing aligned with open science principles.
2. KSGAAL provides a series of open educational resources to teach Arabic: https://www.youtube.com/@Arabic4world_kids.
<https://www.youtube.com/@arabic4world>.
<https://ksaa.futurex.sa>.

4.5 Have there been any initiatives in your country to support science communication in line with open science values and principles with a view to the dissemination of scientific knowledge to researchers of different disciplines, decision-makers, and the public at large?

Yes there have been several Initiatives that potentially support science communication in line with open science values and principles, among which:

1. Open-Access Publishing and Institutional Repositories
 - a) National efforts have focused on making publicly funded research freely accessible through open-access journals, digital repositories, and institutional archives (<https://www.neom.com/en-us>, [SAIP Open Access & IP Support](#))
 - b) Several universities and research institutions have adopted open-access policies, ensuring scientific publications and research data are available to the global community ([KAUST Repository](#), [KAUST Research Data Management](#), [KACST Open Publications](#)).
2. Development of Open Educational Resources (OERs)
 - a) The expansion of OER platforms has provided freely accessible educational content in Arabic and English, supporting researchers, students, and the public ([OERx Official Platform](#), [FutureX Official Site](#)).
 - b) Open science training programs and online courses have been integrated into these platforms to enhance scientific literacy and interdisciplinary knowledge sharing ([Saudi Digital Library](#), [Ain Learning Platform](#)).
3. Digital Science Communication and AI-Powered Learning
 - a) Interactive AI-powered platforms have been launched to support digital learning, research collaboration, and open data analysis ([Elm Research Initiatives](#)).
 - b) Digital environments for scientific experimentation and AI-driven learning tools facilitate interdisciplinary engagement in open science ([NeLC AI Learning Tools](#), [KAWL on X \(Twitter\)](#)).
4. Science-Policy Engagement and Public Awareness
 - a) Dedicated efforts have been made to bridge the gap between scientific research and decision-making, ensuring that policy-relevant scientific findings are communicated effectively to government stakeholders ([NLP](#)

2023 Conference, ArabicNLP 2024 - Sponsors, <https://nelc.gov.sa/en/media/news/2939>).

- b) Science communication campaigns, webinars, public lectures, and industry events have been organized to raise awareness of open science principles among policymakers and the general public ([KAUST Public Talks & Open Science](#)).

5. National and International Collaboration on Open Science

- a) Saudi institutions have established global partnerships with international organizations such as UNESCO, the OECD, and the World Bank to align national science communication efforts with global standards ([KACST https://www.kacst.edu.sa/](#), [NEOM Research & Science Policy](#) and [SAIP Open Research Support](#)).
- b) National agencies have hosted scientific conferences, trend foresight discussions, and interdisciplinary knowledge-sharing initiatives to foster global collaboration in open science ([RDIA Open Science Policy](#), And [KACST https://www.kacst.edu.sa/](#))(<https://x.com/rdia?lang=en>) (<https://www.linkedin.com/company/research-development-and-innovation-authority-rdia/>).

5. FOSTERING A CULTURE OF OPEN SCIENCE AND ALIGNING INCENTIVES FOR OPEN SCIENCE

5.1 Have there been any initiatives at the national or institutional level, e.g. by universities and/or research-performing institutions, to review the existing research assessment and evaluation processes to align them with open science principles and values or are they foreseen by the end of 2025?

Yes, Saudi Arabia recognizes the imperative of establishing a more effective, dynamic, and responsive framework to align with its future aspirations for enhancing the current research assessment and evaluation processes within its national Research, Development, and Innovation (RDI) ecosystem. To achieve these goals, Saudi Arabia has proactively initiated several initiatives aimed at advancing the research assessment and evaluation process, including, but not limited to, the following:

1. NeLC incorporates open science principles into its research assessment and evaluation processes, shifting away from traditional journal-based metrics like impact factors. Instead, it prioritizes the quality, impact, and reusability of research outputs. This approach is exemplified by its focus on industrial research that informs decision-making, with findings and reports directly shared with policymakers to support evidence-based strategies (<https://nelc.gov.sa/en/>).
2. SDAIA is the competent authority in Saudi Arabia concerned with data and AI including big data, and has partnered with Saudi universities to initiate a review of research assessment criteria to incorporate open science principles:
 - a) Alignment with Open Access Policies: efforts are underway to ensure that research outputs prioritize open-access publication, promoting transparency and accessibility of scientific knowledge.
 - b) Integration of Open Science Metrics: new evaluation metrics are being introduced to assess the extent of data sharing, use of open-source tools, and adherence to FAIR principles (Findable, Accessible, Interoperable, Reusable) in research projects.
 - c) Collaboration with Higher Education Institutions: partnership with universities such as King Saud University to focus on revising promotion tenure guidelines to recognize contributions to open science practices (<https://sdaia.gov.sa/en/SDAIA/Pages/default.aspx>).

3. NEOM's ERI aims to establish a world-class education, research, and innovation ecosystem that inspires and empowers top talent. As part of its commitment to Open Science, ERI-supported research findings must be published in peer-reviewed journals and presented at conferences and workshops. Additionally, public engagement, including contributions to white papers and trend reports, is encouraged to influence policymaking. The review and assessment process plays a crucial role in ensuring these objectives are met. NEOM's annual performance cycle evaluates progress against set goals, including Open Science-related activities. This comprehensive review process actively monitors the delivery of Open Science commitments, ensuring alignment with NEOM's broader research and innovation vision.
5. UQU has initiated a review of its research assessment and evaluation processes to better reflect open science principles and values.
 - a) Key aspects under consideration include:
 - Emphasis on Open Access Outputs: Recognizing publications in open-access journals or repositories in research evaluations.
 - Data Sharing and FAIR Principles: Rewarding researchers for making datasets and methodologies openly available and reusable.
 - Broadening Metrics Beyond Journal Impact Factor: Exploring alternative metrics like societal impact, research reproducibility, and engagement with open science practices.
 - Recognition of Collaborative Efforts: Acknowledging interdisciplinary and team-based research contributions.
 - b) Specific Actions Taken:
 - Workshops and consultations with stakeholders (faculty, students, and administrators) to identify gaps in current research assessment frameworks.
 - Piloting new evaluation criteria in specific departments or programs that prioritize open science outputs.
 - c) Future Plans (By End of 2025):
 - Adoption of a comprehensive policy to integrate open science principles into university-wide research assessment criteria.

- Development of training programs for evaluators and faculty to ensure fair implementation of open science-aligned assessment methods.
 - Establishment of metrics and tools to measure the impact and quality of open science contributions.
- d) Collaboration and Benchmarking:
- Partnering with national and international organizations to align university practices with global standards.
- e) Monitoring and Continuous Improvement:
- Periodic review of the revised research assessment processes to ensure alignment with evolving open science practices and values.
 - Collecting feedback from researchers and evaluators to refine and improve implementation.

In addition, Journals owned and published by UQU are open-access journals(<https://uqu.edu.sa/en/jll/112722>).

6. Endorsement of the Global Research Council's Responsible Research Assessment (RRA). The GRC RRA framework is founded on a comprehensive array of initiatives, principles, and guidelines aimed at shaping methodologies, systems, and cultures pertinent to research assessment and evaluation. These elements are organized around five core pillars, each offering essential guidance for cultivating a responsible, transparent, and inclusive research environment:

- a) Research assessment shapes research culture.
- b) Diverse R&I sectors create high-quality research and impact. This should be supported in research assessments.
- c) Carefully considered and formed research assessment criteria and processes foster a healthy R&I system:
 - Research assessment criteria should be broad and use indicators that assess what is valued.
 - Processes play a key role in creating a research assessment system that is fit for purpose.
 - Funder self-evaluation should be an essential part of RRA.

- d) Global approaches to RRA are pivotal to high-quality research and impact but they must be cognisant of local context, culture, and language.
- e) Funders must steward the system by acting and exerting the power they hold to influence other stakeholders.

By endorsing the RRA principles, main pillars, and the 2021 UNESCO Recommendation on Open Science within Saudi Arabia's R&D ecosystem, several universities and research-performing institutions have actively undertaken a review of their existing research assessment and evaluation processes.

5.2 Have specific clauses that recognize open science practices, including sharing, collaborating, and engaging with other researchers and societal actors beyond the scientific community, or dialogue with other knowledge systems, been incorporated in the career evaluation and progression systems or are foreseen by the end of 2025?

Yes, RDIA, as the governing body for the research, development, and innovation ecosystem in Saudi Arabia, is formulating strategies, regulations, and guiding frameworks to enhance the ecosystem. As part of this effort, RDIA has integrated global best practices and experiences, including the principles and foundations of Open Science, into career evaluation and progression systems for research professionals. Alongside this effort, several research institutes and universities have also incorporated Open Science principles into their regulations for researcher evaluation and career progression. For example:

1. KACST is revising its researcher's career evaluation and progression systems, to align with its new role as Saudi Arabia's national laboratory. This new system will adopt a holistic approach to research assessment, focusing on a wide range of contributions, from traditional academic outputs to real-world impacts on industry and society. Key aspects of the new system include:
 - a) Holistic Evaluation: Moving beyond traditional metrics like publication counts, the new framework will emphasize patents, prototypes, and industry partnerships. The focus will be on knowledge creation, innovation, and societal applications.
 - b) Diverse Outputs: In addition to peer-reviewed publications, KACST will give significant importance to tangible outputs like licensing agreements, revenue from intellectual property, and the commercialization of research.

- c) Collaboration and Leadership: Researchers' involvement in the broader scientific community (e.g. conferences, mentoring, editorial work) and their ability to collaborate across disciplines and institutions will be key factors in their evaluation.
 - d) Commercialization and Industry Engagement: The framework will incentivize researchers to pursue innovations that can be commercialized, focusing on successful technology transfers and market-driven solutions.
 - e) Balanced Assessments: A combination of qualitative and quantitative metrics will ensure transparency and fairness, acknowledging the unique strengths of each researcher while fostering innovation and societal impact.
2. The internal regulations of the KAWLA include provisions that ensure the expansion of the scope of open science, including what is included in the policy of digital access to manuscripts and publications at KAWLA, by providing auxiliary models of the information containers owned by the complex through its digital repository, to view one of the forms of implementation: <https://dar.kawla.gov.sa/ar>
 3. JazanU has incorporated specific clauses into its career evaluation and progression systems to recognize and promote open science practices. These clauses emphasize sharing, collaboration, and engagement with researchers and societal actors beyond the scientific community, as well as dialogue with other knowledge systems. Full implementation of these measures is anticipated by the end of 2025. Promotion criteria now include not only published research but also academic contributions such as participation in seminars, conferences, and open discussion panels. Faculty job performance is also evaluated based on a range of academic, research, and community engagement activities, aligning with the principles of open science.

5.3 Have there been any initiatives in your country to recognize and reward open science practices or are they foreseen by the end of 2025?

YES, several universities and research institutes have implemented mechanisms to recognize and reward open science practices, as follows.

1. The TVTC has implemented initiatives to recognize and reward open science practices, with plans to expand these efforts by the end of 2025. These initiatives aim to encourage researchers to adopt transparency, collaboration, and inclusiveness in their work.

a) Current Initiatives:

- Excellence Rewards: Researchers are eligible for monetary rewards based on their contributions to open-access publications, collaborative projects, and the sharing of research data. These rewards aim to incentivize researchers to align their work with open science principles.
- Governor's Award for Applied Research: This prestigious award recognizes outstanding research efforts that embody open science practices, including interdisciplinary collaboration and societal engagement. The award is managed by the organization in collaboration with top national research entities such as KACST. [Governor's Award for Applied Research.](#)

b) Planned Enhancements by 2025:

- New Recognition Programs: Introduction of additional awards for projects that demonstrate significant impact through open science practices, such as engaging societal stakeholders or producing openly accessible datasets.
- Awareness Campaigns: Conducting workshops and seminars to raise awareness about the benefits of participating in open science initiatives and the rewards associated with them.

These initiatives reflect the organization's commitment to fostering a culture of open science and encouraging researchers to embrace its principles. (<https://tvtc.gov.sa/En/About/Pages/AboutUs.aspx>).

2. NeLC has launched the "eLearning Innovation Award" to recognize national achievements and innovative practices in enhancing human capabilities through eLearning. This award encourages eLearning providers and practitioners to develop creative solutions and implement best practices. A key criterion for the award is the organization's efforts in resource sharing (<https://nelc.gov.sa/en/>).
3. MOENERGY plans to launch an Accelerator Program in 2025 to provide a platform for entrepreneurs to compete, with winning ideas rewarded through grants to support the commercialization of promising innovations.
4. JazanU has introduced initiatives to acknowledge and reward open science practices. These efforts include recognizing contributions to collaborative research, promoting open-access publishing, and valuing active involvement in community engagement and knowledge-sharing activities. Looking ahead, the university plans to expand these rewards, ensuring greater recognition for open science contributions by 2025 (<https://www.jazanu.edu.sa/en/colleges/sci/rdu>).

5. The Saudi Electronic University promotes open-access publishing but does not yet have a formalized open-science framework addressing all key elements. The SEU sets [Scientific Research Incentives](#) to support open-access publication fees.
6. Elm Company developed an Internal policy for paper publication rewards in alignment with the organization's strategic goals to promote research excellence. This policy was adopted in 2022 and updated in 2023. The Key areas/sections of the policy include:
 - a) Reward Structure: Incentives for faculty and researchers who publish in reputable journals and conferences.
 - b) Eligibility Criteria: Guidelines on the types of publications and research outputs that qualify for rewards.
 - c) Application Process: Procedures for submitting publications for consideration and claiming rewards.
 - d) Evaluation Metrics: Standards for assessing the impact and quality of research publications.
7. MiyahThon is an initiative by SWA that brings together innovators, engineers, researchers, and entrepreneurs from various fields to collaborate in developing innovative technologies and solutions aimed at addressing the significant challenges in the water sector. MiyahThon is the perfect place to turn ideas into tangible solutions that contribute to the future of water. MiyahThon is proud to collaborate with a diverse group of strategic partners, including industry leaders, technology innovators, and sustainability advocates. These partners provide valuable expertise, resources, and mentorship, empowering participants to develop cutting-edge water solutions. (<https://www.swa.gov.sa/en/news/43>).

5.4 Have there been any unintended negative consequences of open science practices reported in your country?

Yes, there have been some unintended adverse consequences associated with open science practices, particularly in relation to research assessment and evaluation. Key challenges include the risks of predatory behaviors, the potential exploitation of research data, and the increased financial burden on researchers, such as high article processing charges. These issues may exacerbate inequalities within the scientific community, making it harder for early-career researchers and smaller institutions to fully engage in open science practices. Moreover, concerns over intellectual property rights and data security have been raised, given the increased openness and sharing of scientific outputs.

Additionally, the shift toward open science practices may complicate traditional research assessment systems, which are still evolving to align with the new frameworks of openness. This transition requires significant resources and changes in academic output metrics, which can cause friction, especially when universities and institutions feel pressure from global rankings that emphasize traditional research outputs like publications and citations.

6. PROMOTING INNOVATIVE APPROACHES FOR OPEN SCIENCE AT DIFFERENT STAGES OF THE SCIENTIFIC PROCESS

6.1 Have there been any initiatives at the national or institutional level that promote innovative, participatory methodological or procedural changes at different stages of the research cycle to increase the openness of the entire scientific process?

Yes, Saudi Arabia is actively pursuing its Research, Development, and Innovation (RDI) aspirations in alignment with the Vision 2030 goals. The country is implementing various initiatives to foster scientific openness and encourage participatory research, including but not limited to the following:

1. Promotion of Pre-prints

Saudi universities actively contribute to open access, demonstrating a strong commitment to transparency in science. This shift has reduced the number of subscription-only articles and significantly increased open-access publications. KAUST leads the way with its Open Access Policy and a repository for preprints, showcasing over 18,000 publications. Other Saudi universities also support the dissemination of early-stage research.

- a) KAUST's Commitment to Open Access, preprints, and Scientific Innovation: KAUST launched its Open Access Policy in 2014, mandating that research outputs by KAUST authors are made freely accessible. The KAUST Repository, an initiative of the University Library, expands the impact of conference papers, technical reports, peer-reviewed articles, preprints, theses, images, data sets, and other research-related works from KAUST. Over the past decade, KAUST has made over 18,000 publications freely available, achieving more than 3.5 million downloads. Through 20 transformative agreements with publishers such as Wiley and AIP, KAUST supports cost-effective open access for its researchers, reflecting its leadership in the Middle East and its high ranking in open-access output among Asian universities. Additionally, since 2015, KAUST Open Access Week has promoted awareness and discussions on open access, preprints, and data sharing, in alignment with international Open Access Week initiatives. Notable events include the 2019 Discussion Panel on Preprints, which emphasized the importance of preprints in accelerating scientific communication and fostering transparency.
- b) Promoting Transparency in Research: King Saud University Journals' Support for Preprints: The journals of King Saud University (KSU), including Science and Computer and Information Sciences, actively support the use of preprints by acknowledging them in their author guidelines. Authors are

encouraged to reference preprints essential to their work, reflecting KSU's commitment to enhancing early-stage research visibility and promoting transparency in the dissemination of scientific findings, while clearly distinguishing between preliminary and peer-reviewed publications.

- c) KAUST and Saudi Universities' Partnership with MDPI's IOAP: KAUST has partnered with MDPI to provide discounts on Article Processing Charges (APCs) for researchers publishing in Multidisciplinary Digital Publishing Institute (MDPI) journals. Through MDPI's Institutional Open Access Program (IOAP), KAUST benefits from flexible agreements that simplify publishing in CC-BY Open Access journals. Other Saudi universities involved in MDPI's IOAP include Al Imam Mohammad Ibn Saud Islamic University, Princess Nourah bint Abdulrahman University (PNU), and the University of Tabuk.

2. Valuing and Sharing of Negative Scientific Results and Associated Data.

Saudi Arabia is undertaking substantial investments to empower scientists in exploring innovative methodologies. During the Kingdom's address at the Science and Technology session of the Society Forum 2023 in Japan, represented by KACST, the significance of high-quality research output was emphasized, highlighting that success is intricately linked to effective evaluation, despite the inherent complexities involved. The address also stressed the importance of publishing and disseminating negative scientific results, as well as those that do not align with anticipated outcomes, recognizing their role in the advancement of scientific knowledge.

- a) KAUST Data Reproducibility Workshop: The KAUST Data Reproducibility Workshop focuses on addressing the challenges of reproducibility in scientific research. It emphasizes the importance of well-documented, structured, and standardized data to ensure that research findings can be consistently replicated. Through interactive sessions, the workshop equips researchers with the tools and best practices needed to enhance the credibility and reliability of their work. This initiative is part of KAUST's commitment to advancing transparency and rigor in the scientific process.
- ## 3. Participatory methods that value inputs from societal actors, e.g. citizen science, crowdsourcing scientific projects

Participatory methods that value inputs from societal actors are becoming increasingly important in Saudi Arabia, particularly in the context of Vision 2030, which aims to diversify the economy and improve the quality of life for citizens.

- a) The National Center for Vegetation Development and Combating Desertification has initiated a Citizen Science program, designed to engage the public in a collaborative effort to preserve natural resources

- and tackle pressing environmental issues. This interactive initiative serves as a model for productive partnerships among various societal sectors. Participants are encouraged to collect and document images of flora, fauna, and other biodiversity, contributing to a rich repository of data gathered by both specialists and enthusiasts across diverse vegetation-rich regions. The program's objectives include aiding researchers and environmental scientists in the identification of new or previously unrecorded plant species, evaluating plant health, assessing the effects of climate change on growth patterns, and compiling comprehensive information on plant diversity. Ultimately, the initiative seeks to enhance understanding and establish a robust database of our natural resources.
- b) **Crowdsourcing for Smart City Planning:** In Jeddah, researchers have developed an innovative crowdsourcing application known as 'SMART JEDDH.' This platform empowers citizens to actively engage in urban development by reporting real-time issues such as traffic congestion, parking shortages, and air quality concerns directly from their smartphones. The data collected through this application is essential for informed decision-making, enabling city planners to identify problem areas, optimize traffic flow, and prioritize infrastructure enhancements. The study advocates the utilization of crowdsourced smartphone applications for data collection, highlighting that the extensive data amassed through public participation is pivotal in advancing the efficacy of smart cities.
 - c) The research conducted by KAUST researchers on oceanic manta rays in the Red Sea utilized data collected from local divers and social media contributions to analyze manta ray populations. By actively engaging the diving community and encouraging public participation in sharing observations, the researchers successfully identified critical habitats, including a potential nursery ground near Sharm El-Sheikh, Egypt. This innovative crowdsourced methodology significantly enhanced the understanding of manta ray demographics and movement patterns, thereby contributing to conservation initiatives and advancing knowledge of marine biodiversity.
 - d) The "Citizen Science: iNaturalist Bioblitz" workshop, conducted in June 2022 at KAUST, aimed to actively engage participants in the documentation of local biodiversity within the campus environment. Utilizing the iNaturalist platform, attendees investigated the coastal plains and mangroves, meticulously recording species and contributing significant data to global biodiversity databases. This workshop not only enhanced participants' appreciation for Saudi Arabia's natural ecosystems but also offered practical learning experiences, facilitated

by professional ecologists who provided expertise in species identification, thereby promoting a collaborative and participatory approach to scientific inquiry.

- e) Saudi Arabia's e-Participation Initiative: Saudi Arabia's e-Participation initiative fosters communication between the government, citizens, and private sector in line with Vision 2030. It enables the public to participate in the decision-making process and to share input on laws, development projects, and government services, as well as petitions to the Shura Council and ideas for solving public and private sector challenges. Citizens can also participate in budget discussions by suggesting national and local projects.
- f) The Global Research Council (GRC) has articulated a Statement of Principles on Public Engagement that underscores the importance of inclusive research practices. This framework advocates for the active involvement of public groups, promoting transparency, knowledge sharing, and alignment with societal challenges to maximize the impact of scientific endeavors. As a key participant in the GRC, Saudi Arabia has expressed its full endorsement of these principles, demonstrating a commitment voluntarily to engaging the public in its research initiatives. This approach aims to ensure that scientific advancements are beneficial to communities and effectively address pressing challenges. Such involvement is pivotal in enhancing Saudi Arabia's research landscape by fostering innovation and inclusivity.

4. Development of Participatory Strategies for Identifying the Needs of Marginalized Communities and Highlighting Socially Relevant Issues to be Incorporated into Research Agendas.

Aligned with Vision 2030's commitment to a vibrant society, participatory methods that amplify diverse voices are becoming integral in Saudi Arabia. These approaches build a strong social infrastructure by honoring cultural traditions, national pride, and modern values while incorporating all citizens and residents in decision-making to address socially relevant issues, fostering inclusivity and enhancing the quality of life.

- a) Community Participation Policy: The Ministry of Human Resources and Social Development (HRSD) has implemented digital participation mechanisms to enhance citizen involvement in decision-making. Through platforms like "Your Voice Matters," HRSD invites all citizens, residents, governmental organizations, and businesses to express their views, share experiences, and provide feedback using tools like comments, appraisals, direct contact, blogs, polls, and questionnaires

available on its electronic portal. This input informs policy and program development, ensuring that diverse voices are heard and promoting inclusive decision-making.

- b) The "We Care About You" initiative: An initiative by HRSD aims to promote a dignified life and social equity across all life stages. Key focuses include women's empowerment, youth engagement, support for people with disabilities to ensure independence, promoting health and safety in workplaces, improving the quality of life for the elderly, and providing social protection and child development programs. The initiative seeks to build an inclusive and supportive society for all.
- c) King Abdulaziz Center for Cultural Communication (KACCC): The center plays a key role in fostering national unity and promoting constructive dialogue in Saudi Arabia, actively incorporating diverse voices. Through the Department of Studies and Research, KACCC conducts rigorous scientific studies on social issues, develops social indicators to measure public attitudes, and implements national surveys to support decision-makers and guide the center's activities. The department also collaborates with experts, publishes research, and develops implicit bias tests to understand issues like workplace bias toward female leadership. These initiatives align with the center's goals of promoting justice, equality, and freedom of expression, all within the framework of Islamic Sharia.
- d) SDAIA's Role in Driving Social Transformation through AI: SDAIA promotes programs such as "AI for Good", leveraging artificial intelligence to tackle societal challenges, including universal healthcare, equitable access to education, and environmental sustainability. Through initiatives like the SDAIA Academy, which equips professionals with AI skills; the Center of Excellence for AI in Health (AIH), which enhances healthcare innovation through AI-driven solutions; and the Global AI Summit (GAIN), where experts collaborate to ensure AI's impact remains positive and transformative, SDAIA underscores the role of AI as a powerful catalyst for social transformation and addressing critical societal needs.
- e) The Elevate Initiative by SDAIA: This initiative aims to address the gender gap in artificial intelligence by empowering over 25,000 women from emerging markets to enter the fields of data and AI over the next five years. This initiative focuses on fostering inclusivity and diversity in the AI sector, providing women with the tools, knowledge, and opportunities to thrive in these transformative fields. It also ensures that women's

perspectives and needs are incorporated into the broader AI research and development agenda.

5. Education and Training Evaluation Commission (ETEC): The School Evaluation and Accreditation Standards (2023) focus on enabling schools to optimize resources and improve performance using standardized evaluation tools. These standards emphasize inclusivity by addressing the needs of diverse learners, including students with disabilities and gifted students, fostering equitable educational opportunities. Promoting Open Innovation Practices.

Saudi Arabia is advancing open innovation to strengthen collaboration and knowledge sharing across sectors. Key initiatives aim to address challenges like limited collaboration and a need for centralized innovation hubs. These efforts foster a more inclusive RDI ecosystem, supporting greater interaction among researchers, institutions, and the public.

- a) RDIA is the leading authority driving collaboration and innovation in Saudi Arabia through many initiatives such as:
 - Open Access Portal: The Open Access Portal connects researchers and innovators with research infrastructure across Saudi Arabia, fostering collaboration and optimizing the use of resources, aligning with open innovation practices.
 - Open Innovation Collaboration Platform: The Open Innovation Collaboration Platform promotes hackathons and crowdsourced solutions, addressing limited collaboration by fostering partnerships and enabling the co-location of key players within the RDI ecosystem, in alignment with Saudi Arabia's mission-oriented RDI strategy
- b) KACST serves as both a reference National Lab, providing public access to expertise, laboratories, and advanced technologies, and as a Science and Technology Park, driving innovation, supporting R&D commercialization, technology transfer, entrepreneurial ventures, and maximizing the value of research assets. KACST's open infrastructure fosters greater cooperation and collaboration across sectors, enabling the exchange of ideas, technologies, and expertise.
- c) The Garage: the innovation hub established through a collaboration between the MCIT, KACST, and the Saudi Federation for Cybersecurity, Programming, and Drones. It serves as a dynamic ecosystem designed to support emerging technology-based startups. It provides the resources needed for growth and success through a unique business model that integrates incubators, accelerators, and comprehensive support programs,

while fostering a collaborative environment for both local and international startups.

- d) The National Technology Development Program (NTDP): NTDP is a national program that contributes to developing the technology ecosystem in the Kingdom and increasing its effectiveness by driving sustainable growth using different interventions and supporting mechanisms complementing efforts made by other stakeholders. The program also plays an essential role in enhancing all enablers that influence the technology market growth, aiming to make Saudi Arabia an attractive regional center for innovation and investment in the Technology sector.
- e) Open Data Platform by SDAIA: Open data is mandated in Saudi Arabia through the Data Sharing Policy and Open Data Policy, which primarily require government entities to share and make data accessible to foster innovation and enhance transparency by publishing high-value datasets in free and reusable formats. Managed by SDAIA, the Open Data Platform by SDAIA serves as a central hub for public access to data from 271 organizations including health, education and training, agriculture and fishing, statistics, and transport and communication. The platform allows researchers, developers, and citizens to explore and use data for various purposes, including research, innovation, and community projects.
- f) SDAIA Hackathons and Challenges to Drive Open Source Innovation: SDAIA hosts national and international hackathons, including the Smartathon to address visual pollution in Saudi cities, the Generative Artificial Intelligence Hackathon to turn high-tech AI ideas into viable businesses, the Medical Data Fusion Hackathon in collaboration with Stanford University to advance healthcare applications. Moreover, the AI Artathon merges art and AI for creative innovation, and the NEOM Challenge, which encourages students to propose AI-driven solutions for NEOM's development. These events focus on developing open-source solutions and fostering collaborative problem-solving by engaging a diverse range of participants.
- g) KAUST's event on the Future of Data Sharing: FAIR Data, FAIRmat, and NOMAD: KAUST actively supports FAIR (Findable, Accessible, Interoperable, Reusable) data principles through events like The Future of Data Sharing: FAIR Data, FAIRmat, and NOMAD. The event emphasizes the importance of structured, standardized, and well-documented data for large-scale studies, enabling researchers to compare thousands of systems, uncover patterns, and understand general laws governing application performance. It also highlights the FAIRmat consortium, which builds FAIR

data infrastructure for condensed-matter physics, and NOMAD, a tool for managing FAIR-compliant materials science data.

- h) **Aramco-KAUST Consortium for Non-metallics and Composites in Energy Applications:** it is a five-year collaborative initiative between industry and academia to drive advancements in energy materials, focusing on non-metallic and composite solutions to reduce energy use and carbon emissions. Aramco, as a founding member and research sponsor, contributes industry expertise gained from its international initiatives, such as the Non-metallic Innovation Center in the UK and the Carbon Hub at Rice University in the US. KAUST provides advanced research capabilities in engineering, AI, and material science, positioning ENERCOMP as a hub for cutting-edge research and development. ENERCOMP also invites additional energy sector partners, creating a platform for knowledge sharing and innovation to develop practical solutions for the energy transition.
- i) **The Fikra Platform by TAM:** "Fikra" is an initiative that supports open innovation by providing a platform where individuals, businesses, and government bodies collaborate to address complex challenges and develop impactful solutions. This approach promotes collective innovation, ensuring solutions reflect diverse perspectives. By connecting sectors, developing local talent, and involving citizens, Fikra strengthens the national innovation ecosystem and accelerates problem-solving.
- j) **The Innovation Center by Monsha'at:** A dynamic hub designed to support and accelerate innovation-driven entrepreneurship in Saudi Arabia. It offers a range of services and resources tailored to help startups and small to medium enterprises (SMEs) turn innovative ideas into successful, scalable businesses. Through specialized programs like accelerators, the center provides mentorship, training, funding opportunities, and access to networks of industry experts. Additionally, the center fosters a collaborative environment where entrepreneurs can connect, share insights, and collaborate on solutions to complex business challenges.
- k) **The NeLC has the AI Sandbox,** an initiative in digital learning designed to foster innovation in education by providing a safe, flexible environment for testing and developing AI solutions. The program focuses on enhancing learning quality, accelerating AI adoption in education, and enabling evidence-based research. It supports fields such as teaching with AI, adaptive digital curriculum development, and skills advancement. The sandbox offers tools, data access, and expert consultations, targeting researchers, developers, institutions, and companies committed to advancing AI in learning.
- l) **Education and Training Evaluation Commission's Standards for Institutional Accreditation:** The Accreditation Standards emphasize research and innovation, effective governance, and community partnerships, aligning with open innovation principles. These standards foster collaboration,

resource optimization, and integration of diverse stakeholders to enhance knowledge sharing and support innovation. This framework positions institutions to drive advancements in research and societal impact.

m) KACARE drives innovation through technology localization, collaborative workshops, and inclusive feedback mechanisms.

- Localization of Technologies Initiative: In its fourth cycle, with 105 applicants and companies participating. This program aims to enhance the local adoption of advanced technologies.
- Collaborative Workshops: Organized to gather proposals and solutions from internal departments and external stakeholders, promoting collective problem-solving.
- Surveys and Feedback Mechanisms: Designed to engage internal and external beneficiaries, incorporating their input into the research and innovation processes.

n) KAIMRC facilitates open innovation through strategic collaborations with national and international partners, addressing pressing scientific and societal challenges through shared expertise and resources. The partnership between AstraZeneca and KAIMRC includes a joint R&D lab in Riyadh for rare disease drug discovery and integration into AstraZeneca's global A. Catalyst Network to foster collaboration, and technology transfer initiatives that enhance local expertise and drive innovation in healthcare. Additionally, KAIMRC hosts the Riyadh Global Medical Biotechnology Summit and partners with Sanofi to develop an AI-driven disease management ecosystem, leveraging shared expertise to deliver impactful solutions.

o) SAIP National Network of IP Support Centers: A national network to enable innovators to obtain technical information in the fields of intellectual property and facilitate their access to related services to encourage the investment of their innovative potential and the development, protection, and management of their intellectual property rights.

p) QU CODE Lab: the CODE Lab at QU, in collaboration with MCIT, stimulates digital entrepreneurship by providing an ecosystem for startups and fostering collaboration. It offers services such as technical support, incubation programs, and a modeling lab to assist in developing and launching digital projects.

q) KFSHRC supports medical research and innovation through the Biobank Center and SPInE Initiatives

- Biobank Center: Provides high-quality biological resources to advance medical research, personalized medicine, and innovative healthcare solutions.

- SPInE Initiative: Integrates artificial intelligence in healthcare to foster innovation, collaboration, and improved patient outcomes.
- r) The Water Technology Innovation and Research Development Institute (WTIIRA):
- WTIIRA Hosts the Water Research Community: Provides a platform for researchers to collaborate on water research and innovation.
 - WTIIRA Hosts the Innovation-Driven Water Sustainability Conference: Organizes conferences to promote sustainable water technology solutions.
 - WTIIRA Engages in Open Innovation Collaborations: Partners with universities, research institutes, and private companies to address water-related challenges.
 - SWIC Hosts Hackathons: Conducts hackathons to encourage innovation and creative problem-solving in water sustainability.
 - SWIC Hosts Deal Days for Mature Technologies: Facilitates deal days to support the commercialization of advanced water technologies.
- s) SASO Research and Studies Department: SASO's initiatives focus on improving the scientific research infrastructure and developing standards and metrics that support open science, emphasizing the importance of making data and scientific results openly and transparently available. Key contributions include:
- Collaboration with academic and industrial institutions to enhance research outcomes.
 - Developing and updating standards aligned with the latest scientific advancements.
 - Facilitating access to research outputs and fostering a culture of scientific research and openness.

7. PROMOTING INTERNATIONAL AND MULTI-STAKEHOLDER COOPERATION IN THE CONTEXT OF OPEN SCIENCE AND WITH VIEW TO REDUCING DIGITAL, TECHNOLOGICAL AND KNOWLEDGE GAPS

7.1 How is your country promoting and facilitating international scientific collaborations on open science as outlined in the 2021 Recommendation on Open Science?

Saudi Arabia is proactively advancing international scientific collaborations in the realm of open science through various initiatives designed to enhance and facilitate these partnerships. These collaborations are structured under agreements and Memoranda of Understanding (MOUs), which provide a robust framework for cooperation among parties, whether on a bilateral or multilateral basis.

1. The National Portal for Open Access to Research Infrastructure, established by RDIA, provides local and international researchers with access to advanced laboratories and research facilities across Saudi universities and research centers.
2. Open Access Policy for Government-Funded Research Infrastructure, ensuring broad accessibility to scientific resources and data for international collaboration.
3. Establishment of the Joint Centers of Excellence in partnership with leading institutions, such as the Center for Complex Engineering Systems with MIT, focusing on sustainable energy, water management, and smart cities.
4. Establishing strategic partnerships to advance emerging technologies and building a robust semiconductor ecosystem, among which, KACST signed (13) local and international partnerships during LEAP 2024.
5. Participating in international scientific organizations to coordinate national science and technology initiatives while reinforcing global partnerships, including co-organizing workshops with international associations such as co-organizing workshops between KACST and the International Associations of Science Parks and Areas of Innovation (IASP) to enhance science and technology park collaborations.
6. Collaborative research agreements with institutions like Harvard Medical School's Brigham and Women's Hospital (BWH), led to the establishment of the Center of Excellence for Biomedicine, which focuses on cutting-edge research in stem cell therapeutics, nanomedicine, and neurological disorders. Additionally, multiple Memorandums of Understanding (MOUs) have been signed with global research institutions and universities to facilitate joint research projects.
7. Promoting open science through international conferences, global initiatives, and educational partnerships. Collaborations with the Online

Learning Consortium (OLC), OECD, GRC, and the World Bank support research and data sharing, while partnerships with edX and Coursera provide flexible learning pathways via the FutureX platform.

8. Engagements with organizations like ICESCO and UNESCO's IRCAI promote e-learning, AI ethics, and data governance.
9. Hosting the Global AI Summit and Open Education Week, further advocating for open-access research and participatory scientific values.
10. Digital repositories that eliminate temporal and spatial constraints will facilitate the dissemination of open knowledge to diverse target groups globally. This accessibility enhances the availability of cultural assets and general knowledge resources, thereby invigorating the knowledge industry and promoting scientific transparency. By adopting this approach, digital repositories project a sophisticated model of digital accessibility that can be shared with pertinent stakeholders. Consequently, it is anticipated that this will foster various forms of structured international scientific collaboration among competent entities by the conclusion of 2025.

Through these initiatives, Saudi Arabia continues to advance open science, ensuring global knowledge accessibility, collaboration, and innovation in line with international best practices.

7.2 Does your country have a strategy or a plan for stimulating cross-border multi-stakeholder collaboration on open science, including with regard to the exchange of best practices, capacity building, and metrics for open science?

Yes, Saudi Arabia has developed a strategic approach to stimulating cross-border multi-stakeholder collaboration on open science, focusing on the exchange of best practices, capacity building, and the development of metrics for open science.

1. Through the Technology Leaders Program (TLP), Saudi Arabia offers training and research opportunities in collaboration with international partners, bridging gaps in national human resources and enhancing technical leadership in key technological fields.
2. As a longstanding member of the International Science Council since 1987, Saudi Arabia plays a pivotal role in coordinating scientific activities within the country while promoting applied research and global scientific collaboration. Saudi Arabia actively participates in global forums and conferences to address advancements and challenges in open science, fostering an environment of knowledge sharing and international cooperation. Capacity-building efforts include organizing workshops and training programs with international stakeholders, as well as partnering with global organizations to ensure access to resources and tools that

support open science initiatives. Additionally, Saudi Arabia is committed to developing measurable indicators aligned with international standards to evaluate the impact and adoption of open science practices.

3. Also, Saudi Arabia is actively expanding its research and innovation networks through regional and international partnerships, including agreements with global AI research centers to foster knowledge exchange. Saudi Arabia supports innovation through initiatives such as the Saudi Water Innovation Center and the Global Prize for Innovation in Desalination, addressing pressing global challenges. Furthermore, it fosters interdisciplinary collaborations to strengthen research efforts, particularly in areas such as water management and sustainability.
4. Additionally, the Reactivating & Rebuilding of Existing Labs Initiative enhances research capabilities by providing funding for equipment upgrades, maintenance, and human resource development. This initiative aims to attract, hire, and train highly skilled researchers and students, expanding Saudi Arabia's research capacities and positioning it as a global leader in innovation and development.
5. KAWLA is working to prepare a number of forms of open infrastructure that primarily aim to encourage multiple stakeholders to share the knowledge they possess, so that it is made available in an open manner that achieves public benefit, regardless of specialization, geographic location, gender, etc.

By integrating these strategies and initiatives, Saudi Arabia is actively promoting international collaboration in open science, facilitating knowledge exchange, and advancing scientific progress on a global scale.

7.3 Is your country involved in any discussion on the establishment of regional and international funding mechanisms for open science?

Yes, Saudi Arabia is actively engaged in establishing regional and international funding mechanisms to support its growth. Aligned with Vision 2030's emphasis on global scientific collaboration and innovation, the Kingdom is working to develop sustainable financial frameworks that enhance resource accessibility, promote knowledge sharing, and foster strategic partnerships. To this end, Saudi Arabia collaborates with international organizations like ICESCO to explore funding opportunities and participates in global forums and conferences to shape sustainable financial models for open science. Additionally, Saudi Arabia advocates for funding initiatives that drive advancements in critical areas such as next-generation desalination technologies and water treatment solutions, facilitating interdisciplinary research in these fields. Through partnerships with global open science organizations, Saudi Arabia continues to expand its funding capabilities, ensuring long-term support for open science initiatives and collaborative research efforts worldwide.

8. GENERAL CONSIDERATIONS

8.1 Are there any specific external causes that may impede the implementation of the 2021 UNESCO Recommendation on Open Science in your country?

Yes, causes that primarily impede the implementation of the 2021 UNESCO Recommendation on Open Science in Saudi Arabia, including but not limited to:

1. Lack of Open Science Regulatory Frameworks: There is an imperative to establish formal policies and frameworks that explicitly advocate for the integration of open science principles into existing research funding mechanisms, in alignment with the 2021 UNESCO Recommendation.
2. Lack of National Alignment: Currently, there is an absence of a national framework or incentives that support private companies in developing funding mechanisms for open science, which complicates the justification and internal implementation of such funding.
3. Limited Awareness: There is a significant gap in the widespread understanding of open science principles among researchers and stakeholders, compounded by challenges in allocating the necessary resources for awareness and training initiatives.
4. Resource Limitations: The allocation of funds specifically for open science initiatives presents financial challenges, as there is a pressing need to prioritize projects that directly enhance the competitive positioning of the private sector in the marketplace.
5. Coordination Issues: It is essential to ensure alignment with both national and international standards while effectively addressing local needs.
6. Proprietary Concerns: There are apprehensions regarding the sharing of proprietary information, which could potentially undermine the market advantage of private sectors within a competitive industry.

ANNEX 1: PARTICIPATED ORGANIZATIONS AND CANDIDATES

	Name	Website	Representatives
Public & Semi-Public			
1)	Al-Jouf University	ju.edu.sa	Ebtsam Alenezy
2)	Center of Legal Research and Studies at the Bureau of Experts	https://clsr.gov.sa	Dr. Saad Abdulaziz Alotaibi
3)	Education & Training Evaluation Commission	https://etec.gov.sa/home	Abdullah Moteb Albugami
4)	Elm Company	https://elm.sa	Dr. Muhammad Saleh AL-Qurishi
5)	Hail University	www.UOH.edu.sa	Abdullah Zaid Saad almankory
6)	Imam Abdulrahman bin Faisal University	https://www.iau.edu.sa/en	Dr. Ahmed Alkuwaiti
7)	Islamic University of Madinah	https://oerx.nelc.gov.sa/iu https://iu.edu.sa/	Mohammed bin Eid Al-Hajili
8)	Jazan University	https://www.jazanu.edu.sa/en	Dr. Turki Mesfer Alqahtani
9)	King Abdulaziz City for Science and Technology (KACST)	https://kacst.gov.sa	Eng. Ibrahim Alshadokhi Dr. Saad Aldoihi Eng. Fayez Alnefaie. Abdurrahman Alsahaby.
10)	King Abdulaziz Complex for Endowment Libraries	https://www.kawla.gov.sa	Eng. Yasser Alofi
11)	King Abdul-Aziz Public Library	https://kapl.org.sa	Abdullah Faisal Assiri
12)	King Abdullah City for Atomic and Renewable Energy - KACARE	https://www.kacare.gov.sa/ar/Pages/default.aspx	Suzan Katamoura
13)	King Abdullah Petroleum Studies and Research Center	www.kapsarc.org	Axel Pierru

14)	King Fahad National Library	https://kfnl.gov.sa/ar/Pages/default.aspx	Abdullah Ateah Alghamdi
15)	King Faisal University	www.kfu.edu.sa	Prof. Fawaz Waselallah Alsaade
16)	King Khalid University (KKU)	www.kku.edu.sa	Dr. Mohammed Maray
17)	King Salman Global Academy for Arabic Language	https://ksaa.gov.sa/en/	Rawan Almatham
18)	King Saud bin Abdulaziz University for Health Science, KAIMRC	https://www.ksau-hs.edu.sa/English/	Dr. Adel Almutiri
19)	Literature, Publishing & Translation Commission	https://lpt.moc.gov.sa/en	Munirah Alharbi
20)	Local Content & Government Procurement Authority	https://lcpa.gov.sa/	Abdullah Ahmed Alsaeed
21)	Ministry of Communication and Information Technology	https://www.mcit.gov.sa/en	Abdulrahman Alharthi
22)	Ministry of Economy and Planning	https://mep.gov.sa/en	Hamza Hussein Bakhtawer
23)	Ministry of Education	www.moe.gov.sa	Dr. Ibrahim Azizi Dr. Bjad Albudairy
24)	Ministry of Energy	https://www.moenergy.gov.sa/ar/Pages/default.aspx	Ahmad Wadea Khayyat
25)	Ministry of Environment, water, and agriculture	MEWA.gov.sa	Abdulaziz Abdullah Alsuwailem
26)	Ministry of Industry and Mineral Resources	https://mim.gov.sa/en/	Sulaiman Fahad Alshubaili
27)	Ministry of Media	www.media.gov.sa	Ministry of Media
28)	MODON – Saudi Authority for Industrial Cities and	MODON.gov.sa	Hamza Al Amer

	Technology Zones		
29)	Museum Commission	https://museums.moc.gov.sa/en	Dr. Taghreed Alsaraj
30)	National Center for Social Studies	https://ncss.gov.sa/	Atheer alqutaimi
31)	National Cybersecurity Authority	https://www.nca.gov.sa	Majed Ali Alsahli
32)	National eLearning Center	https://nelc.gov.sa/en	Fahad Alenezi
33)	National Transformation Program Center	www.vision2030.gov.sa/en/explore/programs/national-transformation-program	Roba Abdualziz AlQarawi
34)	NEOM	https://www.neom.com/en-us	Prof. Donal Bradley
35)	Prince Saud AL Faisal Institute for Diplomatic Studies	www.ids.gov.sa	Adel Hassan Alomrani
36)	Prince Sultan Defense Studies and Research Center	https://www.psdsarc.org.sa/	Homoud Alrwais
37)	Qassim University	qu.edu.sa	Abdullah Saleh Aljohani
38)	Research, Development, and Innovation Authority (RDIA)	https://rdia.gov.sa/	Prof. Mohammed Alshamsi Dr. Ibrahim Alrammah Dr. Amani Alhibshi
39)	Saudi Authority for Intellectual Property	SAIP.gov.sa	Turki Alfagheeh
40)	Saudi Data and AI Authority	https://sdaia.gov.sa	Dr. Najla Altwaijry
41)	Saudi Electronic University	seu.edu.sa	Dr. Mohammed Kutbi
42)	Saudi National Institute of Health	Saudi NIH	Ghiath Alahmad
43)	Saudi Standards, Metrology, and Quality Organization	https://www.saso.gov.sa/en/pages/default.aspx	Dr.Majed alharthy
44)	Saudi Water Authority	www.swa.gov.sa	AAI-Sahary

45)	Taibah University	https://www.taibahu.edu.sa/pages/en/home.aspx	Dr. Ranya Jumah Aloufi
46)	Technical and Vocational Training Corporation (TVTC)	https://tvtc.gov.sa/ar/About/Pages/default.aspx	Reem Abdulaziz AlMugbel
47)	The Custodian of the Two Holy Mosques King Salman bin Abdulaziz Al Saud Complex for the Prophet's Hadith	Under Development	Eng. Mohammed Alrehaili
48)	The national center of research and development for sustainable	https://estidamah.gov.sa/en/Pages/default.aspx	Saif Fuhaid Alharbi
49)	Umm Al-Qura University	uqu.edu.sa	Meshari Ahmed Alsharif
50)	University of Bishah	https://www.ub.edu.sa/ar/Pages/default.aspx	Dr. Bakri Awaji
51)	University of Jeddah	www.uj.edu.sa	Aala Abdulaziz alzahrany
Private			
52)	Federation of Saudi Chambers	https://fsc.org.sa/	Abdullau Hassan Alomran
53)	King Abdullah University of Science and Technology	https://www.kaust.edu.sa	Dr. Thibaut LERY
54)	SAMI-AEC	https://www.aecl.com	Eng. Asim Altuwayjiri
55)	STC	www.stc.com.sa	JAMAL AHMAD ALSHAHRI
Civil Society			
56)	Scientific Research & Innovation Association	https://sria-sa.net	Dr. Abdullah Alaqal, Dr. Abdulazizi Alammam, Dr. Qamrah Alsubaie
Non-profit Org			
57)	King Abdulaziz and His	https://www.mawhiba.org/en/pages/default.aspx	Fahad S. Alfaiz

	Companions Foundation for Giftedness and Creativity		
58)	King Faisal Specialist Hospital and Research Centre	https://www.kfshrc.edu.sa/	Dr. Khaled Saleh Alhadyan
59)	King Salman Science Oasis	https://ks-scienceoasis.org.sa/	Professor Ibtisam Badhrees

King Abdulaziz City for Science and Technology ([KACST](#))
Riyadh_The Kingdom of Saudi Arabia